Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 1

Basics of Bayesian inference

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **key idea** you need;
- a step-by-step **solution**;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch01/code/ch01.R`.

Sai Sridhar · October 20, 2026

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Answers at a glance

1	$c = e^{1/\theta}/\theta$ ($\theta > 0$): a shifted exponential.
2	A point has zero width $\Rightarrow$ zero area under the PDF.
3	Valid PDF; $\mathrm{E}X = \frac{a+b}{2}$, $\mathrm{Var}X = \frac{(b-a)^2}{12}$.
4	Gamma; Normal; Poisson; Binomial(100); Beta / Exponential / Normal; Binomial(1001).
5	$\mathrm{Gamma}(25/3,\ 5/3)$ (shape, rate).
6	X_1: $(.30, .35, .35)$; X_2: $(.45, .55)$; *not* independent.
7	$X_1 \sim \mathrm{N}(0,1)$; $X_1 \mid X_2 \sim \mathrm{N}(\rho x_2,\ 1-\rho^2)$.
8	Conditionals centred at 0, wider as $\lvert x_2\rvert$ grows: uncorrelated but dependent.
9	$270/484 \approx 0.558$.
10	$0.570/0.578 \approx 0.986$.
11	(a) 0.10 (b) 1.
12	Aleatoric: chaotic small-scale weather. Epistemic: model parameters / sparse observations.
13	Posterior $\propto$ prior $\times\ \theta^k(1-\theta)^{3-k}$; "the" returns the prior.
14	$\mathrm{Prob}(Y{=}0) = 0.804$; $\mathrm{Prob}(Y{=}4) = 0.039$; $\mathrm{Prob}(U{=}0 \mid Y) = 0.995$ if $Y{=}0$, 0 if $Y{=}4$.
15	$\mathrm{Prob}(X_1{=}0 \mid X_2{=}0) = \frac{1-\theta}{1-\theta+\theta(1-\lambda)^n}$; need $n = 28$.
16	$\lambda \mid \mathbf{y} \sim \mathrm{Gamma}(179, 6)$: mean 29.8, 95% CI $(25.6, 34.4)$.
17	ρ: mode 0.85, mean 0.78, 95% CI $(0.49, 0.92)$.
18	Classify as deforested when NDVI < 0.459.
19	$n \mid Y \sim Y + \mathrm{Poisson}(5(1-\theta))$.
20	$\mathrm{Beta}(3,9)$: mean 0.25, SD 0.12; ET $(0.060, 0.518)$, HPD $(0.041, 0.484)$; PPD 0.876.
21	(a) 1 (b) $1/4$ (c) 1 (d) $3/4$.
22	Only Harden shows strong evidence of worse clutch shooting; robust to the prior.
23	(a) 0.993 (c) decreases to 0.410 (d) increases to ≈ 1 (e) attribute if $Y \geq 2$.
24	$p \approx 0.042$; $\mathrm{Prob}(\mathcal{H}_o \mid Y) \approx 0.043$.
25	VE $\approx 93.5\%$, 95% CI $(89.2\%, 96.7\%)$.

Exercise 1.1 — Normalising constant of a shifted exponential

Exercise 1 (exact text from the book)

If X has support $X \in \mathcal{S} = [1, \infty]$, find the constant c (as a function of θ) that makes $f(x) = c\exp(-x/\theta)$ a valid PDF.

In plain words

We are told the *shape* of the density, $e^{-x/\theta}$, but only on $x \geq 1$. Find the number c that makes the total area under the curve equal to one.

Key idea

A function f is a valid PDF on $\mathcal{S}$ when (i) $f(x) \geq 0$ for all $x \in \mathcal{S}$ and (ii) $\int_{\mathcal{S}} f(x)\,\mathrm{d}x = 1$. "Find c" always means: integrate the kernel, then set $c = 1/(\text{integral})$.

Solution

Step 1 — the integral must be finite.

For $\int_1^\infty e^{-x/\theta}\,\mathrm{d}x$ to converge we need $\theta > 0$; for $\theta \leq 0$ the integrand does not decay and no c works.

Step 2 — integrate the kernel.

$$\int_1^\infty e^{-x/\theta}\,\mathrm{d}x = \Big[-\theta e^{-x/\theta}\Big]_1^\infty = 0 - \big(-\theta e^{-1/\theta}\big) = \theta e^{-1/\theta}.$$

Step 3 — solve for c.

$c \cdot \theta e^{-1/\theta} = 1 \Rightarrow c = \dfrac{e^{1/\theta}}{\theta}$. Since $c > 0$, $f(x) \geq 0$ as well.

Sanity check.

Plugging back, $f(x) = \frac{1}{\theta} e^{-(x-1)/\theta}$ for $x \geq 1$. This is an **Exponential distribution with mean θ shifted to start at 1**, i.e. $X = 1 + Z$ with $Z \sim$ Exponential (mean θ). That makes sense: cutting off the region below 1 and re-normalising the memoryless exponential just shifts it.

Answer

$$c = \frac{e^{1/\theta}}{\theta}, \qquad \theta > 0, \qquad \text{so} \qquad f(x) = \frac{1}{\theta}\, e^{-(x-1)/\theta},\ x \geq 1.$$

Takeaway & common mistakes

- Don't forget the lower limit: on $[0, \infty)$ the answer would be $c = 1/\theta$; the extra $e^{1/\theta}$ comes from starting at 1.
- State the parameter restriction ($\theta > 0$); a "constant" that only exists for some θ must say so.

Exercise 1.2 — Why $\text{Prob}(X = x) = 0$ for a continuous variable

Exercise 2 (exact text from the book)

Argue that for continuous random variable X (e.g., normal), $\text{Prob}(X = x)$ must be zero for any single value of x.

In plain words

Explain why a continuous random variable (with a PDF) has zero chance of landing exactly on any one specific number.

Key idea

For a continuous variable, probabilities are *areas* under the PDF: $\text{Prob}(a \le X \le b) = \int_a^b f(x)\,dx$. A single point is an interval of width zero, so it has zero area.

Solution

Argument 1 — squeeze an interval down to a point.

For any $\varepsilon > 0$ the event $\{X = x\}$ is contained in $\{x - \varepsilon \le X \le x + \varepsilon\}$, so

$$0 \;\le\; \text{Prob}(X = x) \;\le\; \int_{x-\varepsilon}^{x+\varepsilon} f(t)\,dt.$$

If f is bounded near x by some M (true for the normal, $M = 1/(\sqrt{2\pi}\,\sigma)$), the right side is at most $2\varepsilon M$. Letting $\varepsilon \to 0$ gives $\text{Prob}(X = x) = 0$. (Without boundedness, the same conclusion follows from continuity of the CDF F: $\text{Prob}(X = x) = F(x) - \lim_{t \uparrow x} F(t) = 0$.)

Argument 2 — counting (intuition by contradiction).

Suppose every point in, say, $[0, 1]$ had probability at least some $\delta > 0$. Pick $N > 1/\delta$ distinct points; the probability of their union would be $N\delta > 1$, which is impossible. So most points must have probability 0; and since the density treats all points of an interval "smoothly", no single point can carry positive mass.

Interpretation.

$f(x)$ is *not* a probability; it is probability *per unit length*. $\text{Prob}(X \in [x, x + \Delta]) \approx f(x)\Delta$, which vanishes as $\Delta \to 0$.

Answer

$\text{Prob}(X = x) = \int_x^x f(t)\,dt = 0$: a single point has zero width, so it carries zero area under the density. Only intervals of positive length can have positive probability.

Takeaway & common mistakes

- A density value can exceed 1 (e.g. $\text{Uniform}(0, 0.1)$ has $f = 10$); this is fine because f is not a probability.
- Consequently $\text{Prob}(X \le x) = \text{Prob}(X < x)$ for continuous X — strict vs. non-strict inequalities don't matter.

Exercise 1.3 — The uniform distribution

Exercise 3 (exact text from the book)

Assume that $X \sim \text{Uniform}(a, b)$ so the support is $\mathcal{S} = [a, b]$, and the PDF is $f(x) = 1/(b-a)$ for any $x \in \mathcal{S}$.

(a) Prove that this is a valid PDF.

(b) Derive the mean and variance of X.

In plain words

Check that a flat density on $[a, b]$ is a real density, then compute its average value and its spread.

Key idea

$\mathrm{E}(X) = \int x f(x)\,\mathrm{d}x$, and $\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$. Useful algebra: $b^2 - a^2 = (b-a)(b+a)$ and $b^3 - a^3 = (b-a)(a^2 + ab + b^2)$.

Solution

(a) Validity.

Non-negativity: $b > a$ so $1/(b-a) > 0$. Total area:

$$\int_a^b \frac{1}{b-a}\,\mathrm{d}x = \frac{b-a}{b-a} = 1. \qquad \checkmark$$

(b) Mean.

$$\mathrm{E}(X) = \int_a^b \frac{x}{b-a}\,\mathrm{d}x = \frac{b^2-a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}.$$

The midpoint — as symmetry demands.

(b) Second moment.

$$\mathrm{E}(X^2) = \int_a^b \frac{x^2}{b-a}\,\mathrm{d}x = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}.$$

(b) Variance.

$$\begin{aligned}\mathrm{Var}(X) &= \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12} \\ &= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12}.\end{aligned}$$

Answer

(a) $f \geq 0$ and $\int_a^b f = 1$, so f is a valid PDF. (b) $\mathrm{E}(X) = \dfrac{a+b}{2}$, $\mathrm{Var}(X) = \dfrac{(b-a)^2}{12}$.

Takeaway & common mistakes

The variance depends only on the *width* $b - a$, not on where the interval sits — shifting a distribution never changes its spread. Memorise $(b-a)^2/12$; it appears constantly (e.g. rounding error).

Exercise 1.4 — Choosing a distribution family

Exercise 4 (exact text from the book)

Select an appropriate family of distributions for the following quantities. Justify your choice.

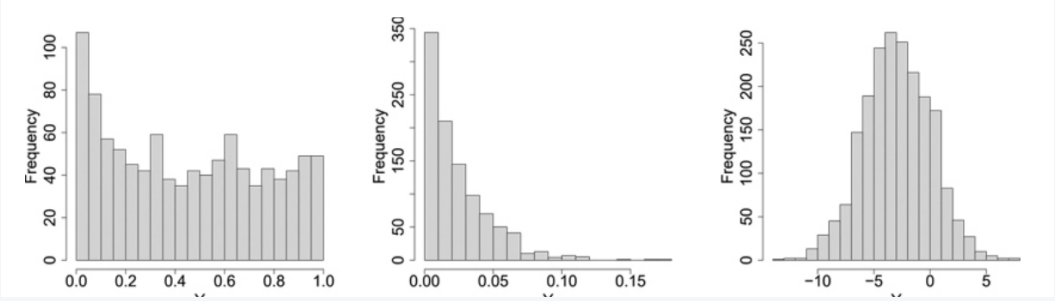
(a) Time between entering Starbucks and having your order.

(b) Heart rate of a randomly selected patient.

(c) Number of hurricanes that make landfall in a given year.

(d) Number of the 100 patients in a clinical trial that experience an adverse event

(e) The data plotted in each of these histograms



(f) The data in this table

Failure	Success
799	202

In plain words

For each quantity, decide what kind of values it can take (yes/no? a count? a count out of a fixed total? a positive number? any real number? a number between 0 and 1?) and match it with a standard distribution.

Key idea

Match the **support** first, then the **shape**:

Support	Natural families
$\{0, 1\}$	Bernoulli
$\{0, 1, \ldots, n\}$ (count out of n)	Binomial
$\{0, 1, 2, \ldots\}$ (unbounded count)	Poisson, negative binomial
$(0, \infty)$ (positive, often skewed)	Exponential, Gamma, Log-normal, Weibull
$(0, 1)$ (a proportion)	Beta
$(-\infty, \infty)$ (symmetric, bell-shaped)	Normal, Student-t

Solution

(a) Waiting time at Starbucks.

A time is positive and continuous and typically right-skewed (most orders are quick, a few take very long). $\Rightarrow$ **Gamma** (or log-normal / Weibull). The exponential is a special case but is a poor fit because it puts its mode at zero — nobody gets coffee in 0 seconds — so a Gamma with shape > 1 is better.

(b) Heart rate.

Heart rates (in beats per minute) are continuous, roughly symmetric and bell-shaped around 60–100, far from zero. ⇒ **Normal**. (Strictly positive, so a log-normal or Gamma is also defensible, but with mean ≈ 75 and SD ≈ 10 the normal puts essentially no mass below 0.)

› (c) Hurricanes per year.

An unbounded count of rare, roughly independent events over a fixed time window. ⇒ **Poisson** (negative binomial if counts are over-dispersed, e.g. due to El Niño years).

› (d) Adverse events among 100 patients.

A count out of a fixed number of independent yes/no trials. ⇒ **Binomial**$(100, \theta)$.

› (e) The three histograms.

- *Left:* values confined to $[0, 1]$, a spike near 0 and a fairly flat, slightly rising pattern up to 1. ⇒ **Beta**(a, b) with $a < 1$ (the spike at 0) and b near or slightly below 1.
- *Middle:* values positive (0 to ≈ 0.15), mode at zero, long right tail. ⇒ **Exponential** (or Gamma with shape ≤ 1).
- *Right:* symmetric, bell-shaped, centred around -2.5 and taking negative and positive values. ⇒ **Normal** (roughly $\mu \approx -2.5$, $\sigma \approx 2.5$).

› (f) Failure/success table.

1,001 binary outcomes with 202 successes. Each outcome is **Bernoulli**(θ), and the total number of successes is **Binomial**$(1001, \theta)$.

Answer

(a) Gamma / log-normal (b) Normal (c) Poisson (d) Binomial$(100, \theta)$
(e) Beta (left), Exponential (middle), Normal (right) (f) Bernoulli trials ⇒ Binomial$(1001, \theta)$

Takeaway & common mistakes

There is rarely one "correct" family; what is graded is the *justification*. Always mention the support (discrete vs. continuous, bounded vs. unbounded) and then the shape (symmetric vs. skewed, mode at the boundary or not).

Exercise 1.5 — Building a prior from a mean and variance

Exercise 5 (exact text from the book)

Expert knowledge dictates that a parameter must be positive and that its prior distribution should have the mean 5 and variance 3. Find a prior distribution that satisfies these constraints.

In plain words

Pick a distribution that lives on positive numbers, then tune its two parameters so that its mean is 5 and its variance is 3.

Key idea

Moment matching. Choose a two-parameter family on $(0, \infty)$, write its mean and variance as functions of the parameters, and solve two equations in two unknowns. The Gamma family is the natural choice: if $\theta \sim \text{Gamma}(a, b)$ (shape a, rate b) then $\text{E}(\theta) = a/b$ and $\text{Var}(\theta) = a/b^2$.

Solution

› Step 1 — set up the equations.

$$\frac{a}{b} = 5, \qquad \frac{a}{b^2} = 3.$$

› Step 2 — divide them.

$\dfrac{a/b}{a/b^2} = b = \dfrac{5}{3}$. Then $a = 5b = \dfrac{25}{3} \approx 8.33$.

› Step 3 — check.

Mean $= \frac{25/3}{5/3} = 5$ ✓; variance $= \frac{25/3}{25/9} = 3$ ✓.

› An equally valid alternative: log-normal.

If $\log\theta \sim \text{Normal}(\mu, \sigma^2)$ then $\text{E}\theta = e^{\mu+\sigma^2/2}$, $\text{Var}\theta = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$, so $\text{Var}/\text{E}^2 = e^{\sigma^2} - 1$:

$$\sigma^2 = \log\left(1 + \tfrac{3}{25}\right) = \log 1.12 \approx 0.1133, \qquad \mu = \log 5 - \tfrac{\sigma^2}{2} \approx 1.5528.$$

Answer

$\theta \sim \text{Gamma}\left(a = \frac{25}{3},\ b = \frac{5}{3}\right)$ (shape–rate) has support $(0, \infty)$, mean 5 and variance 3. (Alternatively $\theta \sim \text{LogNormal}(1.5528,\ 0.1133)$.)

Takeaway & common mistakes

- Two constraints pin down a two-parameter family, but *different families* still give different priors (different tails). The answer is not unique — say so.
- A normal prior with mean 5 and variance 3 is *wrong* here: it puts $\text{Prob}(\theta < 0) = \Phi(-5/\sqrt{3}) \approx 0.002 > 0$, violating positivity.
- Watch the parameterisation: in the shape–*scale* form, the scale is $1/b = 0.6$.

Exercise 1.6 — Marginal and conditional distributions from a table

Exercise 6 (exact text from the book)

X_1 and X_2 have joint PMF

x_1	x_2	$\text{Prob}(X_1 = x_1, X_2 = x_2)$
0	0	0.15
1	0	0.15
2	0	0.15
0	1	0.15
1	1	0.20
2	1	0.20

(a) Compute the marginal distribution of X_1.

(b) Compute the marginal distribution of X_2.

(c) Compute the conditional distribution of $X_1|X_2$.

(d) Compute the conditional distribution of $X_2|X_1$.

(e) Are X_1 and X_2 independent? Justify your answer.

In plain words

From the joint probability table, find the probabilities of each variable on its own, the probabilities of one variable once we know the other, and check whether knowing one changes the other.

Key idea

- **Marginal** = sum the joint over the other variable: $\text{Prob}(X_1 = x_1) = \sum_{x_2} \text{Prob}(x_1, x_2)$.
- **Conditional** = joint ÷ marginal: $\text{Prob}(X_1 = x_1 \mid X_2 = x_2) = \text{Prob}(x_1, x_2)/\text{Prob}(X_2 = x_2)$.
- **Independence** $\iff$ joint = product of marginals for *every* cell (equivalently: every conditional equals the marginal).

Solution

It helps to lay the table out as a 2×3 grid with row and column totals:

	$x_1 = 0$	$x_1 = 1$	$x_1 = 2$	$\text{Prob}(X_2 = x_2)$
$x_2 = 0$	0.15	0.15	0.15	**0.45**
$x_2 = 1$	0.15	0.20	0.20	**0.55**
$\text{Prob}(X_1 = x_1)$	**0.30**	**0.35**	**0.35**	1

› (a) Marginal of X_1 (column sums).

$\text{Prob}(X_1 = 0) = 0.30,\ \text{Prob}(X_1 = 1) = 0.35,\ \text{Prob}(X_1 = 2) = 0.35.$

› (b) Marginal of X_2 (row sums).

$\text{Prob}(X_2 = 0) = 0.45,\ \text{Prob}(X_2 = 1) = 0.55.$

› (c) $X_1 \mid X_2$ (divide each row by its total).

	$x_1 = 0$	$x_1 = 1$	$x_1 = 2$
$X_2 = 0$	$\frac{0.15}{0.45} = \frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$X_2 = 1$	$\frac{0.15}{0.55} = \frac{3}{11} \approx 0.273$	$\frac{0.20}{0.55} = \frac{4}{11} \approx 0.364$	$\frac{4}{11} \approx 0.364$

› (d) $X_2 \mid X_1$ (divide each column by its total).

	$x_2 = 0$	$x_2 = 1$
$X_1 = 0$	$\frac{0.15}{0.30} = \frac{1}{2}$	$\frac{1}{2}$
$X_1 = 1$	$\frac{0.15}{0.35} = \frac{3}{7} \approx 0.429$	$\frac{4}{7} \approx 0.571$
$X_1 = 2$	$\frac{3}{7} \approx 0.429$	$\frac{4}{7} \approx 0.571$

Each row of each conditional table sums to 1 — a quick check.

› (e) Independence.

Check one cell: $\text{Prob}(X_1 = 0, X_2 = 0) = 0.15$, but $\text{Prob}(X_1 = 0)\text{Prob}(X_2 = 0) = 0.30 \times 0.45 = 0.135 \neq 0.15$. One failing cell is enough. Equivalently, the conditional distribution of X_1 changes with X_2 ($\frac{1}{3}$ vs. $\frac{3}{11}$ for $x_1 = 0$).

Answer

(a) X_1: $(0.30, 0.35, 0.35)$ on $(0, 1, 2)$. (b) X_2: $(0.45, 0.55)$ on $(0, 1)$.
(c) $X_1|X_2 = 0$: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$; $X_1|X_2 = 1$: $(\frac{3}{11}, \frac{4}{11}, \frac{4}{11})$.
(d) $X_2|X_1 = 0$: $(\frac{1}{2}, \frac{1}{2})$; $X_2|X_1 = 1$ and $X_2|X_1 = 2$: $(\frac{3}{7}, \frac{4}{7})$.
(e) **Not independent**: $0.15 \neq 0.30 \times 0.45 = 0.135$.

Takeaway & common mistakes

To *prove* independence you must check every cell; to *disprove* it one counter-example suffices. The dependence here is mild: $X_2 = 1$ makes $X_1 \geq 1$ slightly more likely.

Exercise 1.7 — The bivariate normal: marginal and conditional

Exercise 7 (exact text from the book)

If (X_1, X_2) is bivariate normal with $\mathrm{E}(X_1) = \mathrm{E}(X_2) = 0$, $\mathrm{Var}(X_1) = \mathrm{Var}(X_2) = 1$, and $\mathrm{Cor}(X_1, X_2) = \rho$:

(a) Derive the marginal distribution of X_1.

(b) Derive the conditional distribution of $X_1 | X_2$.

In plain words

Start from the joint bell-shaped density of two standard normals with correlation ρ. Show that (a) on its own, X_1 is standard normal, and (b) once we know $X_2 = x_2$, X_1 is normal with a shifted mean and a smaller variance.

Key idea

Complete the square in the exponent. Any expression of the form $\exp\{-(z-m)^2/(2v)\}$ integrates to $\sqrt{2\pi v}$ over $z \in \mathbb{R}$, and as a function of z it is the kernel of $\mathrm{Normal}(m, v)$.

Solution

The joint density.

With unit variances and correlation ρ ($|\rho| < 1$),

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\right\}.$$

Complete the square in x_2.

$$x_1^2 - 2\rho x_1 x_2 + x_2^2 = (x_2 - \rho x_1)^2 + (1-\rho^2)x_1^2.$$

(Expand $(x_2 - \rho x_1)^2 = x_2^2 - 2\rho x_1 x_2 + \rho^2 x_1^2$ and add $(1-\rho^2)x_1^2$ to check.) Therefore

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{x_1^2}{2}\right\} \exp\left\{-\frac{(x_2 - \rho x_1)^2}{2(1-\rho^2)}\right\}.$$

(a) Integrate out x_2.

The last factor is a normal kernel in x_2 with mean ρx_1 and variance $1-\rho^2$, so it integrates to $\sqrt{2\pi(1-\rho^2)}$:

$$f(x_1) = \int f(x_1, x_2)\, \mathrm{d}x_2 = \frac{\sqrt{2\pi(1-\rho^2)}}{2\pi\sqrt{1-\rho^2}} e^{-x_1^2/2} = \frac{1}{\sqrt{2\pi}} e^{-x_1^2/2}.$$

This is exactly the $\mathrm{Normal}(0, 1)$ density — the correlation disappears completely from the marginal.

(b) Divide joint by marginal.

By symmetry (swap the roles of x_1 and x_2 above) $f(x_2)$ is standard normal and

$$f(x_1 \mid x_2) = \frac{f(x_1, x_2)}{f(x_2)} = \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-x_2^2/2} \exp\left\{-\frac{(x_1 - \rho x_2)^2}{2(1-\rho^2)}\right\}}{\frac{1}{\sqrt{2\pi}} e^{-x_2^2/2}} = \frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left\{-\frac{(x_1 - \rho x_2)^2}{2(1-\rho^2)}\right\}.$$

Answer

(a) $X_1 \sim \text{Normal}(0, 1)$. (b) $X_1 \mid X_2 = x_2 \sim \text{Normal}\big(\rho x_2,\ 1 - \rho^2\big)$.

Takeaway & common mistakes

- The conditional mean ρx_2 is a **regression line** through the origin with slope ρ — this is where linear regression comes from.
- The conditional variance $1 - \rho^2$ is *smaller* than 1 and does not depend on x_2: learning X_2 always helps, more so when $|\rho|$ is large.
- When $\rho = 0$ the conditional equals the marginal: for normals, uncorrelated $\Rightarrow$ independent (not true in general — see Exercise 8!).

Exercise 1.8 — Uncorrelated but dependent

Exercise 8 (exact text from the book)

Assume (X_1, X_2) have support $X_1, X_2 \in \mathbb{R}$ and bivariate PDF

$$f(x_1, x_2) = \frac{1}{2\pi}\left(1 + x_1^2 + x_2^2\right)^{-3/2}.$$

(a) Plot the conditional distribution of $X_1 | X_2 = x_2$ for $x_2 \in \{-3, -2, -1, 0, 1, 2, 3\}$ (preferably on the same plot).

(b) Do X_1 and X_2 appear to be correlated? Justify your answer.

(c) Do X_1 and X_2 appear to be independent? Justify your answer.

In plain words

For this joint density, look at how the distribution of X_1 changes as we fix X_2 at different values. Is there a linear trend (correlation)? Is there any dependence at all?

Key idea

- $f(x_1 \mid x_2) \propto f(x_1, x_2)$ as a function of x_1: we can read off the shape and then normalise.
- **Correlation** measures only the linear relationship (does the *centre* of $X_1|X_2$ move with x_2?). **Independence** requires the *whole* conditional distribution not to depend on x_2 — centre, spread, shape.

Solution

› Step 1 — marginal of X_2.

Write $a^2 = 1 + x_2^2$. Using $\int_{-\infty}^{\infty}(a^2 + t^2)^{-3/2}\,\mathrm{d}t = 2/a^2$ (substitute $t = a\tan u$),

$$f(x_2) = \frac{1}{2\pi}\cdot\frac{2}{1 + x_2^2} = \frac{1}{\pi(1 + x_2^2)},$$

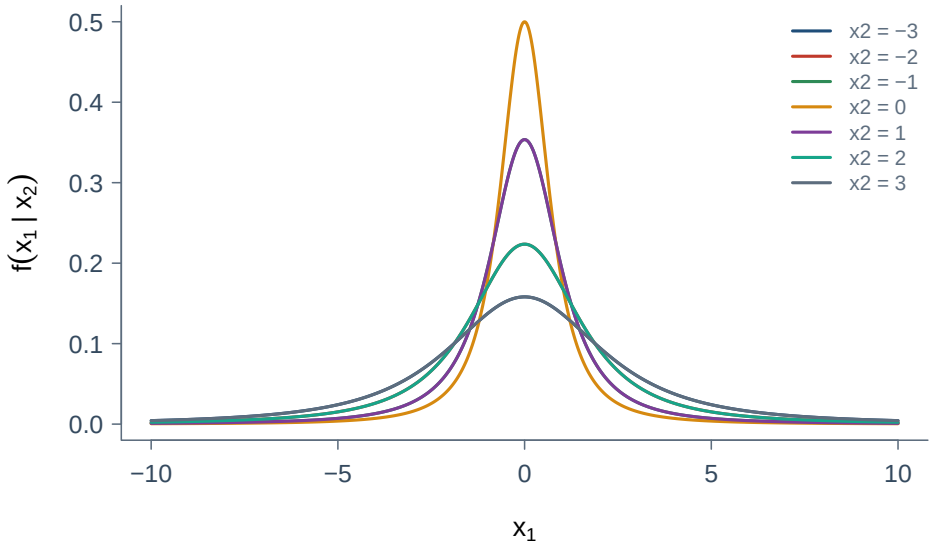
i.e. X_2 (and by symmetry X_1) is **standard Cauchy**. This density is the bivariate t distribution with 1 degree of freedom.

› Step 2 — the conditional in closed form.

$$f(x_1 \mid x_2) = \frac{f(x_1, x_2)}{f(x_2)} = \frac{1 + x_2^2}{2}\left(1 + x_2^2 + x_1^2\right)^{-3/2} = \frac{1}{2a}\left(1 + \frac{x_1^2}{a^2}\right)^{-3/2}, \qquad a = \sqrt{1 + x_2^2}.$$

This is a Student-t with 2 degrees of freedom, centred at 0 and scaled by $a/\sqrt{2}$: $X_1 \mid X_2 = x_2 \overset{d}{=} \sqrt{\frac{1+x_2^2}{2}}\,T_2$. The peak height is $1/(2a)$.

› (a) The plot.



All seven curves are centred at 0. The curve for $x_2 = 0$ is the tallest and narrowest (peak 0.5); for $|x_2| = 3$ the peak drops to 0.158 and the curve is about 3 times wider. The curves for $\pm x_2$ coincide.

› (b) Correlated?

No linear relationship. The conditional centre is 0 for every x_2, and $f(x_1, x_2) = f(-x_1, x_2)$, so there is no tendency for large X_2 to come with large (or small) X_1. Technically the correlation is not even defined, because Cauchy marginals have no finite variance — but there is no trend that a correlation could capture.

› (c) Independent?

No. Independence would require $f(x_1 \mid x_2)$ to be the same for every x_2. Instead the *spread* of X_1 grows with $|x_2|$: knowing $|X_2|$ is large tells us $|X_1|$ is likely large too. Formally, $f(0,0) = \frac{1}{2\pi} \approx 0.159$ while $f_1(0) f_2(0) = \frac{1}{\pi^2} \approx 0.101$.

Answer

(a) See the figure: all conditionals are symmetric about 0 and get wider as $|x_2|$ grows ($X_1|x_2$ is a scaled t_2 with scale $\sqrt{(1 + x_2^2)/2}$). (b) They do **not** appear correlated — the centre never moves. (c) They are **not independent** — the spread changes with x_2.

Takeaway & common mistakes

Uncorrelated ≠ independent. Correlation is blind to relationships in the variance ("heteroscedastic" dependence). Exercise 7 showed the special normal case where the two coincide; this exercise is the standard counter-example.

Exercise 1.9 — Car thefts in the Triangle

Exercise 9 (exact text from the book)

According to insurance.com, the 2017 auto theft rate was 135 per 10,000 residents in Raleigh, NC compared to 214 per 10,000 residents in Durham/Chapel Hill, NC. Assuming Raleigh's population is twice as large as Durham/Chapel Hill and a car has been stolen somewhere in the triangle (i.e., one of these two areas), what the probability it was stolen in Raleigh?

In plain words

Raleigh has a lower theft *rate* but more people. Given a theft happened somewhere in the region, how likely is it that it was in Raleigh?

Key idea

Bayes' rule: $\text{Prob}(R \mid \text{theft}) = \dfrac{\text{Prob}(\text{theft} \mid R)\text{Prob}(R)}{\text{Prob}(\text{theft} \mid R)\text{Prob}(R) + \text{Prob}(\text{theft} \mid D)\text{Prob}(D)}$. The *prior* is the population share; the *likelihood* is the theft rate.

Solution

Prior.

Raleigh is twice as big: $\text{Prob}(R) = \frac{2}{3}$, $\text{Prob}(D) = \frac{1}{3}$.

Likelihood.

$\text{Prob}(\text{theft} \mid R) = 0.0135$, $\text{Prob}(\text{theft} \mid D) = 0.0214$.

Posterior.

$$\text{Prob}(R \mid \text{theft}) = \frac{0.0135 \cdot \frac{2}{3}}{0.0135 \cdot \frac{2}{3} + 0.0214 \cdot \frac{1}{3}} = \frac{2(135)}{2(135) + 214} = \frac{270}{484} \approx 0.558.$$

Intuition with counts.

Imagine 10,000 people in Durham/Chapel Hill and 20,000 in Raleigh: Raleigh would see 270 thefts and Durham/Chapel Hill 214; 270 of the 484 thefts are in Raleigh.

Answer

$\text{Prob}(\text{Raleigh} \mid \text{car stolen}) = 270/484 \approx \mathbf{0.558}$.

Takeaway & common mistakes

A lower rate does not mean fewer events. The larger prior weight (population) outweighs the lower likelihood (rate). Ignoring the prior — answering $135/(135 + 214) = 0.387$ — is the classic *base-rate fallacy*.

Exercise 1.10 — Spam filter

Exercise 10 (exact text from the book)

Say 60% of the emails that arrive in your inbox are spam. Your account correctly flags spam emails as spam with probability 95% and incorrectly flags other emails as spam with probability 2%. Given that an email is flagged as spam, what is the probability it is actually a spam email?

In plain words

If the filter says "spam", how much should we trust it?

Key idea

Bayes' rule with prior $\text{Prob}(S) = 0.6$, sensitivity $\text{Prob}(F \mid S) = 0.95$ and false-positive rate $\text{Prob}(F \mid S^c) = 0.02$.

Solution

› Total probability of a flag.

$\text{Prob}(F) = 0.95(0.6) + 0.02(0.4) = 0.570 + 0.008 = 0.578.$

› Posterior.

$$\text{Prob}(S \mid F) = \frac{0.570}{0.578} \approx 0.9862.$$

› Counts.

Of 1,000 emails, 600 are spam (570 flagged) and 400 are not (8 flagged). Of the 578 flagged emails, 570 are spam.

Answer

$\text{Prob}(\text{spam} \mid \text{flagged}) = 0.570/0.578 \approx \mathbf{0.986}$.

Takeaway & common mistakes

Here the base rate (60%) is high, so a flag is very reliable. If only 1% of emails were spam, the same filter would give $\text{Prob}(S \mid F) = 0.0095/(0.0095 + 0.0198) \approx 0.32$ — the prior matters enormously.

Exercise 1.11 — Commute times and conventions

Exercise 11 (exact text from the book)

Your daily commute is distributed uniformly between 15 and 20 minutes if there is no convention downtown. However, conventions are scheduled for roughly 1 in 4 days, and your commute time is distributed uniformly from 15 to 30 minutes if there is a convention. Let Y be your commute time this morning.

(a) What is the probability that there was a convention downtown given $Y = 18$?

(b) What is the probability that there was a convention downtown given $Y = 28$?

In plain words

From how long your commute took, infer whether there was a convention today.

Key idea

Bayes' rule with a **continuous** observation: replace probabilities of the data by *densities*: $\text{Prob}(C \mid Y = y) = \dfrac{f(y \mid C)\text{Prob}(C)}{f(y \mid C)\text{Prob}(C) + f(y \mid C^c)\text{Prob}(C^c)}$.

Solution

Prior: $\text{Prob}(C) = \frac{1}{4}$. Likelihoods: $f(y \mid C) = \frac{1}{15}$ on $[15, 30]$; $f(y \mid C^c) = \frac{1}{5}$ on $[15, 20]$ and 0 otherwise.

› (a) $Y = 18$ — both scenarios are possible.

$$\text{Prob}(C \mid Y = 18) = \frac{\frac{1}{4} \cdot \frac{1}{15}}{\frac{1}{4} \cdot \frac{1}{15} + \frac{3}{4} \cdot \frac{1}{5}} = \frac{1/60}{1/60 + 9/60} = \frac{1}{10}.$$

› (b) $Y = 28$ — impossible without a convention.

$f(28 \mid C^c) = 0$, so the denominator equals the numerator: $\text{Prob}(C \mid Y = 28) = 1$.

Answer

(a) $\text{Prob}(C \mid Y = 18) = \mathbf{0.10}$ (b) $\text{Prob}(C \mid Y = 28) = \mathbf{1}$

Takeaway & common mistakes

In (a) the posterior (0.10) is *lower* than the prior (0.25): an 18-minute commute is 3 times more likely without a convention (density $\frac{1}{5}$ vs. $\frac{1}{15}$). In (b) the data rule one hypothesis out entirely. Every value $Y \in [15, 20]$ gives the same answer 0.1, because both likelihoods are flat there.

Exercise 1.12 — Aleatoric vs. epistemic uncertainty

Exercise 12 (exact text from the book)

Consider the problem of short-term weather forecasting. Describe (i) a potential source of aleatoric uncertainty and (ii) a potential source of epistemic uncertainty.

In plain words

Give one example of randomness in weather that no amount of extra knowledge could remove, and one example of uncertainty that comes from our lack of knowledge and could be reduced.

Key idea

Aleatoric uncertainty ("dice") is inherent variability in the outcome even if the model and its parameters were known exactly. **Epistemic** uncertainty ("knowledge") comes from not knowing the parameters/model; it shrinks with more data. In Bayesian terms: aleatoric lives in the likelihood $f(Y \mid \theta)$, epistemic in the posterior $p(\theta \mid Y)$.

Solution

(i) Aleatoric.

The atmosphere is chaotic: tiny, unmeasurable turbulent fluctuations determine whether a convective shower forms over *this particular* neighbourhood in the next hour. Even with a perfect model and perfect parameters, whether rain falls at a specific point is genuinely random from our point of view — e.g. "40% chance of rain" when tomorrow's large-scale conditions are known.

(ii) Epistemic.

Uncertainty about the forecast model itself and its inputs: the parameters of the numerical weather model (e.g. how cloud micro-physics is parameterised), or the current state of the atmosphere in regions with sparse weather stations (over oceans). More observations (satellites, buoys) and better calibration reduce this.

Answer

(i) Aleatoric: chaotic small-scale variability (e.g. exactly where a thunderstorm cell forms). (ii) Epistemic: imperfectly known model parameters / initial conditions due to limited observations — reducible with more data.

Takeaway & common mistakes

A quick test: "Would collecting more data make this uncertainty go away?" Yes ⇒ epistemic; no ⇒ aleatoric. The posterior predictive distribution (Section 1.5 of the book) combines both.

Exercise 1.13 — A Bayesian spell checker

Exercise 13 (exact text from the book)

For this problem pretend we are dealing with a language with a six-word dictionary

$$\{\text{fun, sun, sit, sat, fan, for}\}.$$

An extensive study of literature written in this language reveals that all words are equally likely except that "for" is α times as likely as the other words. Further study reveals that:

i. Each keystroke is an error with probability θ.

ii. All letters are equally likely to produce errors.

iii. Given that a letter is typed incorrectly it is equally likely to be any other letter.

iv. Errors are independent across letters.

For example, the probability of correctly typing "fun" (or any other word) is $(1-\theta)^3$, the probability of typing "pun" or "fon" when intending to type is "fun" is $\theta(1-\theta)^2$, and the probability of typing "foo" or "nnn" when intending to type "fun" is $\theta^2(1-\theta)$. Use Bayes' rule to develop a simple spell checker for this language. For each of the typed words "sun", "the", "foo", give the probability that each word in the dictionary was the intended word. Perform this for the parameters below:

(a) $\alpha = 2$ and $\theta = 0.1$.

(b) $\alpha = 50$ and $\theta = 0.1$.

(c) $\alpha = 2$ and $\theta = 0.95$.

Comment on the changes you observe in these three cases.

In plain words

Someone typed a (possibly misspelled) 3-letter word. Combine (1) how common each dictionary word is and (2) how likely the typo is, to get the probability of each intended word.

Key idea

$$\underbrace{\text{Prob}(\text{word } w \mid \text{typed } t)}_{\text{posterior}} \propto \underbrace{\text{Prob}(w)}_{\text{prior}} \times \underbrace{\theta^{k(w,t)}(1-\theta)^{3-k(w,t)}}_{\text{likelihood}},$$

where $k(w,t)$ is the number of letter positions in which w and t differ. Prior: $\text{Prob}(\text{for}) = \frac{\alpha}{5+\alpha}$ and $\text{Prob}(w) = \frac{1}{5+\alpha}$ for the other five words.

Solution

› Step 1 — count the mismatches $k(w,t)$.

typed	fun	sun	sit	sat	fan	for
sun	1	0	2	2	2	3
the	3	3	3	3	3	3
foo	2	3	3	3	2	1

› Step 2 — a useful simplification.

The likelihood is $(1-\theta)^3\, r^k$ with $r = \theta/(1-\theta)$. The factor $(1-\theta)^3$ cancels in Bayes' rule, so

$$\text{Prob}(w \mid t) = \frac{\pi_w\, r^{k(w,t)}}{\sum_{w'} \pi_{w'}\, r^{k(w',t)}}.$$

For $\theta = 0.1$, $r = 1/9$: every extra typo makes a word 9 times less likely. For $\theta = 0.95$, $r = 19$: every extra typo makes a word 19 times *more* likely!

› Worked example: (a), typed "sun".

Weights $\pi_w r^k$ with $\pi \propto (1,1,1,1,1,2)$ and $r = 1/9$: fun $1/9$, sun 1, sit/sat/fan $1/81$ each, for $2/729$. Total $= 1 + 0.1111 + 0.0370 + 0.0027 = 1.1509$. So $\text{Prob}(\text{sun} \mid \text{sun}) = 1/1.1509 = 0.869$, etc.

› Step 3 — all results

(from `ch01.R`):

case	typed	fun	sun	sit	sat	fan	for
(a) $\alpha = 2, \theta = 0.1$	sun	0.097	**0.869**	0.011	0.011	0.011	0.002
	the	0.143	0.143	0.143	0.143	0.143	**0.286**
	foo	0.049	0.006	0.006	0.006	0.049	**0.885**
(b) $\alpha = 50, \theta = 0.1$	sun	0.091	**0.822**	0.010	0.010	0.010	0.056
	the	0.018	0.018	0.018	0.018	0.018	**0.909**
	foo	0.002	0.000	0.000	0.000	0.002	**0.995**
(c) $\alpha = 2, \theta = 0.95$	sun	0.001	0.000	0.024	0.024	0.024	**0.926**
	the	0.143	0.143	0.143	0.143	0.143	**0.286**
	foo	0.017	**0.322**	**0.322**	**0.322**	0.017	0.002

› Step 4 — comments.

- **"the"** differs from *every* word in all 3 letters, so the likelihood is identical for all words and the **posterior equals the prior**: $\frac{\alpha}{5+\alpha}$ for "for" and $\frac{1}{5+\alpha}$ for the others, in every case. Uninformative data ⇒ no update.
- **(a) vs. (b):** increasing α only shifts mass toward "for". With a single typo the data still dominate ("sun" stays 0.82), but when the data are weak ("the") or ambiguous the prior takes over (0.909 for "for").
- **(c)** $\theta = 0.95$**:** the typist almost always mistypes, so a *matching* letter is evidence *against* a word. The checker now prefers words that differ most from what was typed: "sun" → "for" (3 mismatches), "foo" → sun/sit/sat (3 mismatches). The model is internally consistent; it just reflects a terrible typist.

Answer

See the table. Posterior $\propto$ prior $\times\, \theta^k(1-\theta)^{3-k}$. "the" always returns the prior; larger α pulls ambiguous inputs toward "for"; with $\theta = 0.95$ the checker favours the words that *disagree* most with the typed word.

Note

Rule iii ("equally likely to be any other letter") strictly implies that a *specific* wrong letter has probability $\theta/25$, giving likelihood $(\theta/25)^k(1-\theta)^{3-k}$. The book's worked example uses $\theta^k(1-\theta)^{3-k}$ instead, so the table above follows the book. Under the strict rule, case (c) becomes much less extreme (e.g. "sun" → sun 0.229, for 0.201, fun 0.174), because now $r = 0.038/0.05 < 1$ and matching letters are again weak evidence *for* a word. `ch01.R` prints both versions.

⚠ Takeaway & common mistakes

This is a toy version of a real noisy-channel spell checker: *language model* (prior) × *error model* (likelihood). When the likelihood is flat, the prior is all you have; when the error model is wrong, the posterior is confidently wrong.

Exercise 1.14 — The zero-inflated Poisson

Exercise 14 (exact text from the book)

If Y follows the Zero-Inflated Poisson (ZIP) distribution, it can be written in terms of two variables $Y = UV$, where $U \sim \text{Bernoulli}(p)$ independent of $V \sim \text{Poisson}(\lambda)$.

(a) Generate and plot a sample of size $n = 10{,}000$ with $p = 0.2$ and $\lambda = 4$.

(b) What is a real-life example of a variable that might follow this distribution? In your example, what are the interpretations of U and V?

(c) What is the probability that $Y = 0$?

(d) What is the probability that $Y = 4$?

(e) What is the probability that $U = 0$ given that $Y = 0$?

(f) What is the probability that $U = 0$ given that $Y = 4$?

In plain words

Y is a count that is forced to be zero whenever a coin flip U comes up 0; otherwise it is a Poisson count. Simulate it, interpret it, and compute some (conditional) probabilities.

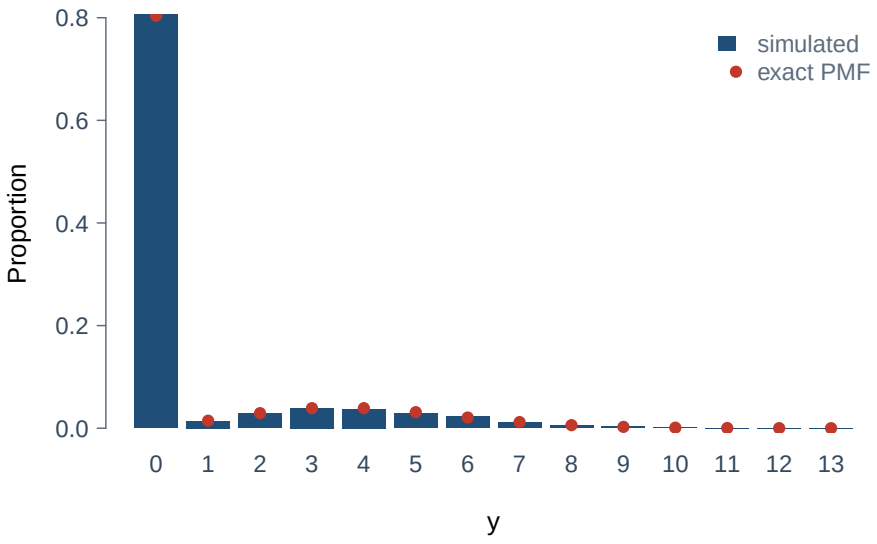
Key idea

There are **two ways** to get $Y = 0$: a "structural" zero ($U = 0$) or a "Poisson" zero ($U = 1, V = 0$). Law of total probability: $\text{Prob}(Y = y) = \text{Prob}(U = 0)\mathbb{1}(y = 0) + \text{Prob}(U = 1)\text{Prob}(V = y)$.

Solution

(a) Simulation.

```
U ← rbinom(10000, 1, 0.2); V ← rpois(10000, 4); Y ← U * V
```



About 80% of the sample is zero; the rest looks like a Poisson(4) scaled down by 0.2. Red dots: exact PMF, matching the simulation.

(b) Example.

Y = number of fish caught by a visitor to a state park. U = indicator that the visitor went fishing at all (most did not), V = number of fish caught *if* fishing. Other examples: cigarettes smoked per day (U = smoker), insurance claims per year (U = ever claims), doctor visits.

(c) $\text{Prob}(Y = 0)$.

$$\text{Prob}(Y=0) = \text{Prob}(U=0) + \text{Prob}(U=1)\text{Prob}(V=0) = 0.8 + 0.2e^{-4} = 0.8 + 0.00366 = \mathbf{0.8037}.$$

(d) $\text{Prob}(Y = 4)$.

Only possible with $U = 1, V = 4$:

$$\text{Prob}(Y=4) = 0.2 \cdot \frac{e^{-4}4^4}{4!} = 0.2(0.1954) = \mathbf{0.0391}.$$

(e) $\text{Prob}(U = 0 \mid Y = 0)$.

Bayes' rule, with $\text{Prob}(Y = 0 \mid U = 0) = 1$:

$$\text{Prob}(U=0 \mid Y=0) = \frac{0.8 \cdot 1}{0.8037} = \mathbf{0.9954}.$$

(f) $\text{Prob}(U = 0 \mid Y = 4)$.

If $U = 0$ then $Y = 0$ with certainty, so $\text{Prob}(Y = 4 \mid U = 0) = 0$ and the posterior is $\mathbf{0}$.

Answer

(c) $0.8 + 0.2e^{-4} \approx 0.8037$ (d) $0.2\,e^{-4}4^4/24 \approx 0.0391$ (e) $0.8/(0.8 + 0.2e^{-4}) \approx 0.9954$ (f) 0
Simulated: $\widehat{\text{Prob}}(Y = 0) = 0.806$, $\widehat{\text{Prob}}(Y = 4) = 0.038$ — consistent.

Takeaway & common mistakes

Note that $p = \text{Prob}(U = 1)$ is the probability of being in the *Poisson* state, so the zero-inflation probability is $1 - p = 0.8$ here. Reading p as "probability of a structural zero" is a common mix-up — check which way the model is written.

Exercise 1.15 — Occupancy with imperfect detection

Exercise 15 (exact text from the book)

Let $X_1 \sim \text{Bernoulli}(\theta)$ be the indicator that a tree species occupies a forest and $\theta \in [0, 1]$ denote the prior occupancy probability. The researcher gathers a sample of n trees from the forest, and X_2 belong to the species of interest. The model for the data is $X_2|X_1 \sim \text{Binomial}(n, \lambda X_1)$ where $\lambda \in [0, 1]$ is the probability of detecting the species given it is present. Give expressions in terms of n, θ and λ for the following joint, marginal and conditional probabilities:

(a) $\text{Prob}(X_1 = X_2 = 0)$.

(b) $\text{Prob}(X_1 = 0)$.

(c) $\text{Prob}(X_2 = 0)$.

(d) $\text{Prob}(X_1 = 0|X_2 = 0)$.

(e) $\text{Prob}(X_2 = 0|X_1 = 0)$.

(f) $\text{Prob}(X_1 = 0|X_2 = 1)$.

(g) $\text{Prob}(X_2 = 0|X_1 = 1)$.

(h) Provide intuition for how (d)-(g) change with n, θ and λ.

(i) Assuming $\theta = 0.5$, $\lambda = 0.1$, and $X_2 = 0$, how large must n be before we can conclude with 95% confidence that the species does not occupy the forest?

In plain words

A species may or may not live in a forest. If it does, each sampled tree is that species with probability λ; if it does not, we never see it. If we sample n trees and see none of the species, how sure can we be it's absent?

Key idea

If $X_1 = 0$ then $X_2 \sim \text{Binomial}(n, 0)$, i.e. $X_2 = 0$ *with certainty*. If $X_1 = 1$, $\text{Prob}(X_2 = 0 \mid X_1 = 1) = (1 - \lambda)^n$ ("missed it n times in a row").

Solution

(a)

$\text{Prob}(X_1 = 0, X_2 = 0) = \text{Prob}(X_1 = 0)\text{Prob}(X_2 = 0 \mid X_1 = 0) = (1 - \theta) \cdot 1 = \mathbf{1} - \theta$.

(b)

$\text{Prob}(X_1 = 0) = \mathbf{1} - \theta$.

(c)

Total probability: $\text{Prob}(X_2 = 0) = (1 - \theta) \cdot 1 + \theta(1 - \lambda)^n = (\mathbf{1} - \theta) + \theta(\mathbf{1} - \lambda)^{\mathbf{n}}$.

(d)

Bayes' rule = (a)/(c):

$$\text{Prob}(X_1 = 0 \mid X_2 = 0) = \frac{1 - \theta}{(1 - \theta) + \theta(1 - \lambda)^n}.$$

(e)

$\text{Prob}(X_2 = 0 \mid X_1 = 0) = \mathbf{1}$ (no species, nothing to find).

(f)

$\text{Prob}(X_1 = 0 \mid X_2 = 1) = \mathbf{0}$: seeing even one tree of the species proves it is present (no false positives in this model).

(g)

$\text{Prob}(X_2 = 0 \mid X_1 = 1) = (\mathbf{1} - \lambda)^{\mathbf{n}}$.

(h) Intuition.

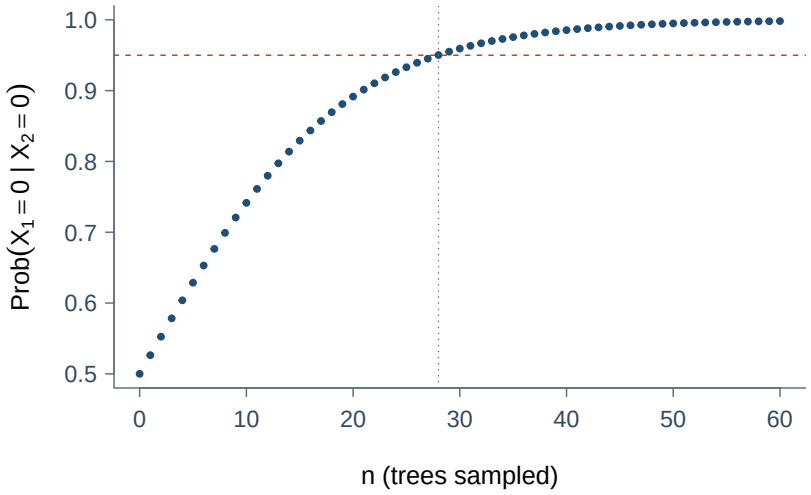
- (d) *increases* with n and λ: the more (and better) we search without finding it, the more convinced we are it is absent; as $n \to \infty$ it tends to 1 (if $\lambda > 0$). It *decreases* with θ: a higher prior belief in presence needs more evidence to overturn.
- (e) and (f) are constant (1 and 0) — they are logical consequences of the model.
- (g) decays geometrically in n, faster for larger λ; it does not depend on θ.

(i) How many trees?

With $\theta = 0.5$, (d) becomes $\dfrac{1}{1 + 0.9^n}$. We need

$$\frac{1}{1+0.9^n} \geq 0.95 \iff 0.9^n \leq \frac{1}{19} \iff n \geq \frac{\log 19}{\log(1/0.9)} = \frac{2.944}{0.1054} = 27.95.$$

Check: $n = 27$ gives 0.9450; $n = 28$ gives 0.9503.



Answer

(a) $1 - \theta$ (b) $1 - \theta$ (c) $1 - \theta + \theta(1 - \lambda)^n$ (d) $\dfrac{1 - \theta}{1 - \theta + \theta(1 - \lambda)^n}$ (e) 1 (f) 0 (g) $(1 - \lambda)^n$

(i) $\mathbf{n = 28}$ trees with no detections.

Takeaway & common mistakes

Absence of evidence is evidence of absence — but only gradually. With a 10% detection rate, 27 empty samples are still not quite enough for 95% certainty. Ignoring imperfect detection (setting $\lambda = 1$) would let you declare absence after a single empty sample.

Exercise 1.16 — Discovery rate of marine bivalve species

Exercise 16 (exact text from the book)

In a study that uses Bayesian methods to forecast the number of species that will be discovered in future years, [47] report that the number of marine bivalve species discovered each year from 2010-2015 was 64, 13, 33, 18, 30 and 20. Denoting Y_t as the number of species discovered in year t and assuming $Y_t|\lambda \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$ and $\lambda \sim \text{Uniform}(0, 100)$, plot the posterior distribution of λ.

In plain words

Six yearly counts come from a Poisson with unknown rate λ. Starting from a flat prior on $(0, 100)$, find and draw the posterior of λ.

Key idea

Posterior $\propto$ likelihood $\times$ prior. With a flat prior the posterior is just the likelihood, restricted to the prior's support. Recognise the **Gamma kernel** $\lambda^{a-1}e^{-b\lambda}$.

Solution

› Step 1 — likelihood.

$$f(\mathbf{y} \mid \lambda) = \prod_{t=1}^{6} \frac{e^{-\lambda}\lambda^{y_t}}{y_t!} \;\propto\; \lambda^{\sum y_t} e^{-6\lambda} = \lambda^{178} e^{-6\lambda}$$

since $64 + 13 + 33 + 18 + 30 + 20 = 178$.

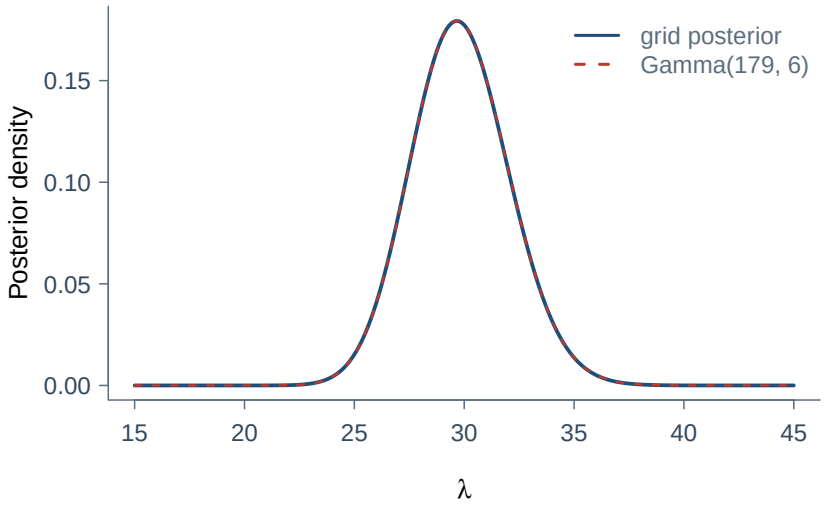
› Step 2 — posterior.

$$p(\lambda \mid \mathbf{y}) \propto \lambda^{178} e^{-6\lambda} \mathbb{1}(0 < \lambda < 100),$$

a $\text{Gamma}(179, 6)$ kernel truncated to $(0, 100)$. The truncation is irrelevant: $\text{Prob}(\lambda > 100) \approx 2 \times 10^{-91}$ under the Gamma. So $\lambda \mid \mathbf{y} \approx \text{Gamma}(179, 6)$.

› Step 3 — summaries.

Mean $179/6 = 29.83$, SD $\sqrt{179}/6 = 2.23$, 95% credible interval $(25.6,\ 34.4)$.



The grid-normalised posterior (solid) and the $\mathrm{Gamma}(179, 6)$ density (dashed) are indistinguishable.

Answer

$\lambda \mid \mathbf{y} \sim \mathrm{Gamma}(179,\ 6)$ (truncated at 100, negligibly): posterior mean ≈ 29.8 species/year, 95% interval $(25.6, 34.4)$. See the plot.

Takeaway & common mistakes

- The posterior SD (2.2) is much smaller than the year-to-year variation in the data (SD ≈ 18). That is fine — it measures uncertainty about the *average* rate — but such large variation (64 vs. 13) suggests the Poisson model (variance = mean) is too narrow; a negative binomial or a time trend would be worth checking.
- The uniform prior is the conjugate $\mathrm{Gamma}(1, 0)$ prior (truncated) — previewing Chapter 2.

Exercise 1.17 — Posterior of a correlation coefficient

Exercise 17 (exact text from the book)

Assume that (X, Y) follow the bivariate normal distribution and that both X and Y have marginal mean zero and marginal variance one. We observe six independent and identically distributed data points: $(-3.3, -2.6)$, $(0.1, -0.2)$, $(-1.1, -1.5)$, $(2.7, 1.5)$, $(2.0, 1.9)$ and $(-0.4, -0.3)$. Make a scatter plot of the data, and, assuming the correlation parameter ρ has a $\text{Uniform}(-1, 1)$ prior, plot the posterior distribution of ρ.

In plain words

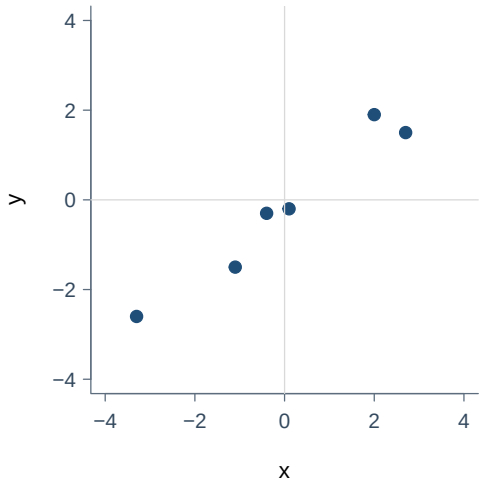
Six points from a standard bivariate normal with unknown correlation ρ. Plot them, and find the posterior of ρ under a flat prior.

Key idea

Only one parameter is unknown, so we can evaluate the unnormalised posterior on a fine **grid** of ρ values and normalise numerically. The likelihood depends on the data only through $S_{xx} = \sum x_i^2$, $S_{yy} = \sum y_i^2$, $S_{xy} = \sum x_i y_i$.

Solution

› Step 1 — scatter plot.



The points lie close to a rising line — a strong positive association.

› Step 2 — likelihood.

Using the density from Exercise 7 for each pair,

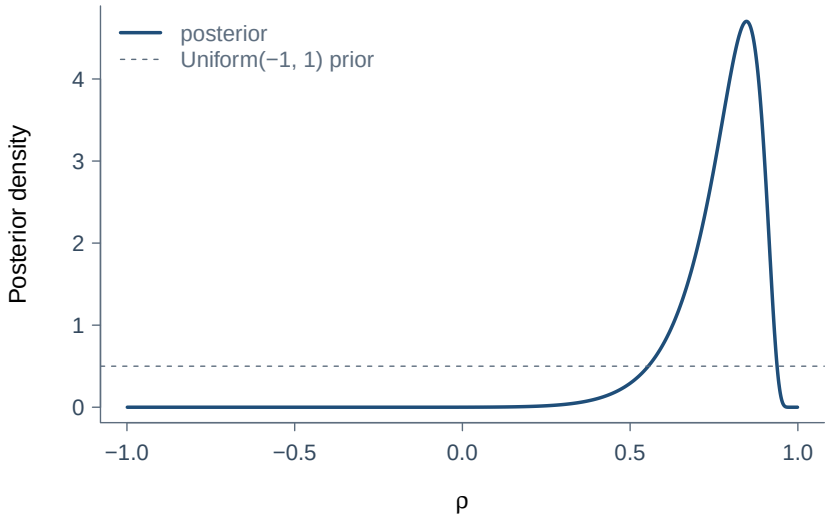
$$f(\text{data} \mid \rho) = \prod_{i=1}^{6} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{ -\frac{x_i^2 - 2\rho x_i y_i + y_i^2}{2(1-\rho^2)} \right\} \propto (1-\rho^2)^{-3} \exp\left\{ -\frac{S_{xx} - 2\rho S_{xy} + S_{yy}}{2(1-\rho^2)} \right\},$$

with $S_{xx} = 23.56$, $S_{yy} = 15.00$, $S_{xy} = 18.18$.

› Step 3 — posterior on a grid.

The prior is flat on $(-1, 1)$, so $p(\rho \mid \text{data}) \propto f(\text{data} \mid \rho)$. Evaluate on 3,999 grid points and divide by the (Riemann) sum.

```
rg ← seq(-0.999, 0.999, length.out = 3999)
ll ← -3*log(1 - rg^2) - (Sxx - 2*rg*Sxy + Syy) / (2*(1 - rg^2))
post ← exp(ll - max(ll)); post ← post / sum(post * diff(rg)[1])
```



› Step 4 — summaries.

Mode 0.85, mean 0.78, 95% credible interval $(0.49,\ 0.92)$, $\text{Prob}(\rho > 0 \mid \text{data}) \approx 1$.

Answer

The posterior of ρ is concentrated on strong positive values: mode ≈ 0.85, mean ≈ 0.78, 95% interval $\approx (0.49, 0.92)$ — see the plot.

Takeaway & common mistakes

The sample correlation of these points is 0.97, but the posterior is centred lower. Why? The model fixes both variances at 1, so $\text{Var}(X - Y) = 2(1 - \rho)$. The observed $\frac{1}{6}\sum(x_i - y_i)^2 = \frac{S_{xx} + S_{yy} - 2S_{xy}}{6} = 0.37$ matches $2(1 - \rho)$ for $\rho \approx 0.82$ — that, not the sample correlation, is what pins ρ down when the variances are known. With $n = 6$ the posterior is also wide and skewed: report the whole distribution, not just a point.

Exercise 1.18 — Classifying deforestation from NDVI

Exercise 18 (exact text from the book)

The normalized difference vegetation index (NDVI) is commonly used to classify land cover using remote sensing data. Hypothetically, say that NDVI follows a $\text{Beta}(25, 10)$ distribution for pixels in a rain forest, and a $\text{Beta}(10, 15)$ distribution for pixels in a deforested area now used for agriculture. Assuming about 10% of the rain forest has been deforested, your objective is to build a rule to classify individual pixels as deforested based on their NDVI.

(a) Plot the PDF of NDVI for forested and deforested pixels, and the marginal distribution of NDVI averaging over categories.

(b) Give an expression for the probability that a pixel is deforested given its NDVI value, and plot this probability by NDVI.

(c) You will classify a pixel as deforested if you are at least 90% sure it is deforested. Following this rule, give the range of NDVI that will lead to a pixel being classified as deforested.

In plain words

Forest pixels have high NDVI; deforested pixels have lower NDVI; only 10% are deforested. For a given NDVI, how likely is the pixel to be deforested, and which NDVI values give at least 90% certainty?

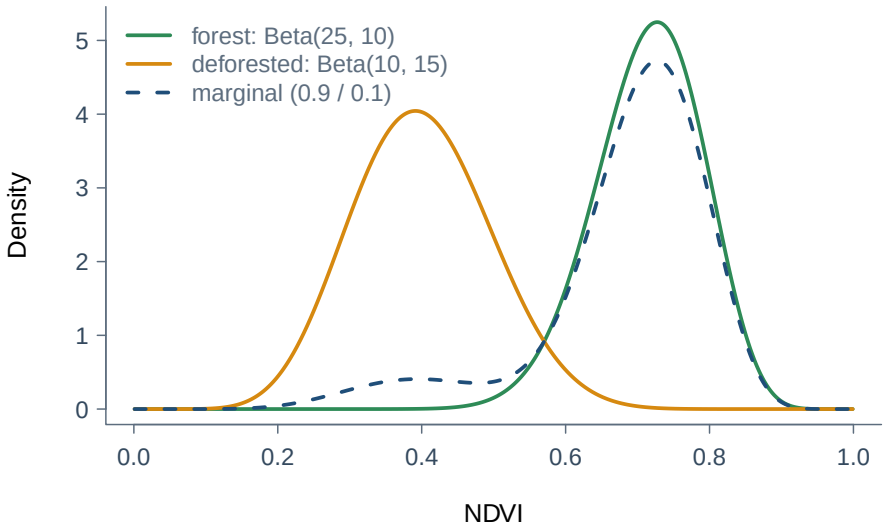
Key idea

A two-class **Bayes classifier**: prior class probabilities $(0.1, 0.9)$ times class-conditional densities, then normalise. The marginal density is the **mixture** $m(x) = 0.9 f_F(x) + 0.1 f_D(x)$.

Solution

(a) Densities.

$\text{Beta}(25, 10)$ has mean $25/35 = 0.71$; $\text{Beta}(10, 15)$ has mean $10/25 = 0.40$.



The marginal (dashed) is dominated by the forest component, with a small shoulder from the deforested pixels at low NDVI.

› (b) Posterior probability of deforestation.

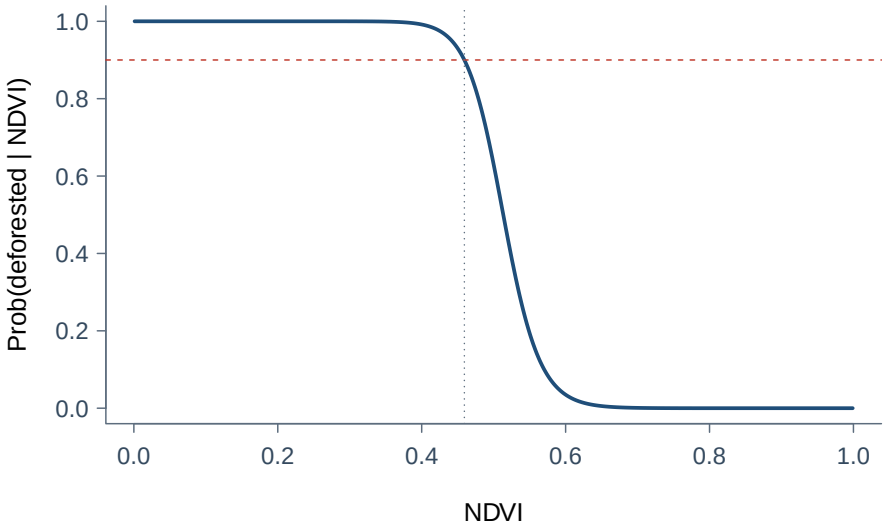
With $f_D = \text{Beta}(10, 15)$ and $f_F = \text{Beta}(25, 10)$ densities,

$$\text{Prob}(D \mid \text{NDVI} = x) = \frac{0.1\, f_D(x)}{0.1\, f_D(x) + 0.9\, f_F(x)}.$$

Writing out the Beta densities, the ratio simplifies nicely:

$$\frac{f_F(x)}{f_D(x)} = \frac{B(10, 15)}{B(25, 10)}\, x^{15}(1 - x)^{-5},$$

which increases in x, so $\text{Prob}(D \mid x)$ *decreases* monotonically in x.



› (c) The 90% rule.

Because $\text{Prob}(D \mid x)$ is decreasing, the rule "$\text{Prob}(D \mid x) \geq 0.9$" is equivalent to $x \leq c$, where c solves $\text{Prob}(D \mid c) = 0.9$. Numerically (`uniroot`), $c = 0.459$.

Answer

(b) $\text{Prob}(D \mid x) = \dfrac{0.1\,\text{Beta}(x; 10, 15)}{0.1\,\text{Beta}(x; 10, 15) + 0.9\,\text{Beta}(x; 25, 10)}$.
(c) Classify a pixel as deforested when $\mathbf{NDVI < 0.459}$ (i.e. NDVI $\in (0,\ 0.459)$).

Takeaway & common mistakes

Error rates of this rule: only 0.1% of forest pixels fall below 0.459 (few false alarms), but only 73% of deforested pixels do — 27% of deforestation is missed. Demanding 90% certainty with a 10% prior is a strict rule; the threshold trades false positives against false negatives.

Exercise 1.19 — How many customers came?

Exercise 19 (exact text from the book)

Let n be the unknown number of customers who visit a store on the day of a sale. The number of customers who make a purchase is $Y|n \sim \text{Binomial}(n, \theta)$ where θ is the known probability of making a purchase given the customer visited the store. The prior is $n \sim \text{Poisson}(5)$. Assuming θ is known and n is the unknown parameter, plot the posterior distribution of n for all combinations of $Y \in \{0, 5, 10\}$ and $\theta \in \{0.2, 0.5\}$, and comment on the effect of Y and θ on the posterior.

In plain words

We only see the number of buyers Y. Infer how many people visited, n, knowing each visitor buys with probability θ.

Key idea

Here the *parameter* n is discrete, so the posterior is a PMF: $p(n \mid Y) \propto \text{Prob}(n)\text{Prob}(Y \mid n)$ for $n = Y, Y+1, \ldots$ (we need $n \geq Y$). A neat closed form comes from **Poisson thinning**.

Solution

› Step 1 — write the posterior.

For $n \geq Y$,

$$p(n \mid Y) \propto \frac{e^{-5}5^n}{n!}\binom{n}{Y}\theta^Y(1-\theta)^{n-Y} \propto \frac{5^n(1-\theta)^{n-Y}}{(n-Y)!} \propto \frac{[5(1-\theta)]^{n-Y}}{(n-Y)!}.$$

› Step 2 — recognise it.

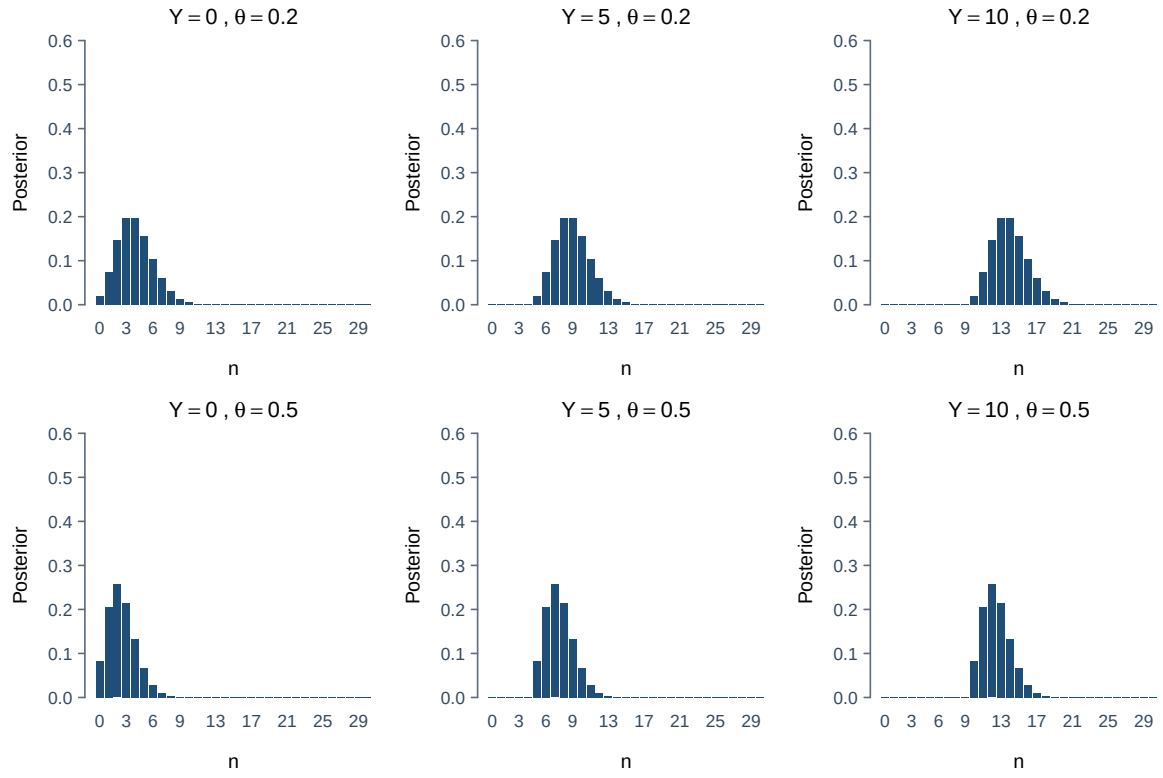
This is a Poisson kernel in $m = n - Y$. Therefore

$$n \mid Y \;=\; Y + M, \qquad M \sim \text{Poisson}\big(5(1-\theta)\big).$$

In words: we know the Y buyers came; the number of *non-buyers* is Poisson with mean $5(1-\theta)$, independent of Y.

› Step 3 — summaries.

$\text{E}(n \mid Y) = Y + 5(1-\theta)$, $\text{Var}(n \mid Y) = 5(1-\theta)$. For $\theta = 0.2$: means $4, 9, 14$ (SD 2.0). For $\theta = 0.5$: means $2.5, 7.5, 12.5$ (SD 1.58).



Comments.

- **Effect of Y:** it simply *shifts* the posterior — the shape is identical for $Y = 0, 5, 10$. Note $Y = 10$ pushes n far above the prior mean of 5: the data override the prior because $n \geq Y$ is a hard constraint.
- **Effect of θ:** larger θ means fewer unseen non-buyers, so the posterior is *tighter* and closer to Y. As $\theta \to 1$, $n \mid Y \to Y$ exactly.
- With $Y = 0$, the posterior ($\text{Poisson}(4)$ or $\text{Poisson}(2.5)$) is shifted *down* from the prior $\text{Poisson}(5)$: no buyers suggests fewer visitors.

Answer

$n \mid Y \sim Y + \text{Poisson}(5(1-\theta))$. Y shifts the posterior right without changing its spread; larger θ makes it narrower and closer to Y. See the six panels.

Takeaway & common mistakes

When you see $\binom{n}{Y}$ with n unknown, don't panic — cancel factorials and look for a familiar kernel. Also remember to restrict the support ($n \geq Y$); forgetting it gives positive probability to impossible values.

Exercise 1.20 — Seedling survival: the beta–binomial model

Exercise 20 (exact text from the book)

Last spring your lab planted ten seedlings, and two survived the winter. Let θ be the probability that a seedling survives the winter.

(a) Assuming a uniform prior distribution for θ, compute its posterior mean and standard deviation.

(b) Assuming the same prior as in (a), compute and compare the equal-tailed and highest density 95% posterior credible intervals.

(c) If you plant another 10 seedlings next year, what is the posterior predictive probability that at least one will survive the winter?

In plain words

2 out of 10 seedlings survived. With a flat prior on the survival probability, summarise the posterior, give two kinds of 95% intervals, and predict next year.

Key idea

$Y \mid \theta \sim \text{Binomial}(n, \theta)$, $\theta \sim \text{Beta}(1, 1) \Rightarrow \theta \mid Y \sim \text{Beta}(Y + 1,\ n - Y + 1)$. For $\text{Beta}(a, b)$: mean $\frac{a}{a+b}$, variance $\frac{ab}{(a+b)^2(a+b+1)}$, and $\text{E}[(1-\theta)^m] = \frac{B(a,b+m)}{B(a,b)}$.

Solution

Posterior.

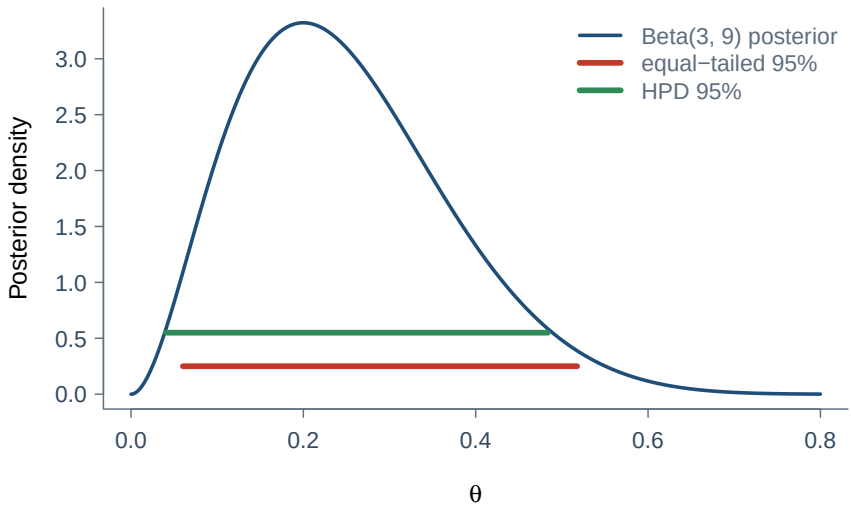
$\theta \mid Y = 2 \sim \text{Beta}(3, 9)$.

(a) Mean and SD.

$$\text{E}(\theta \mid Y) = \frac{3}{12} = 0.25, \qquad \text{SD} = \sqrt{\frac{3 \cdot 9}{12^2 \cdot 13}} = \sqrt{\frac{27}{1872}} = 0.120.$$

(b) Intervals.

Equal-tailed: 2.5% and 97.5% quantiles: `qbeta(c(.025,.975),3,9)` $= (0.060,\ 0.518)$, width 0.458.
HPD: the shortest interval with 95% probability; equivalently the density is equal at both ends. We minimise the width $F^{-1}(F(\ell) + 0.95) - \ell$ over ℓ: HPD $= (0.041,\ 0.484)$, width 0.443. (Check: density at both ends $= 0.585$.)



The posterior is right-skewed, so the HPD interval shifts *left* toward the mode (0.2) and is about 3% shorter.

› (c) Posterior predictive.

Let Y^* be next year's survivors out of 10. Then

$$\text{Prob}(Y^* \geq 1 \mid Y) = 1 - \text{Prob}(Y^* = 0 \mid Y) = 1 - \text{E}\big[(1-\theta)^{10} \mid Y\big] = 1 - \frac{B(3,19)}{B(3,9)}.$$

Using $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$,

$$\frac{B(3,19)}{B(3,9)} = \frac{\Gamma(19)\Gamma(12)}{\Gamma(22)\Gamma(9)} = \frac{11 \cdot 10 \cdot 9}{21 \cdot 20 \cdot 19} = \frac{990}{7980} = 0.1241,$$

so $\text{Prob}(Y^* \geq 1 \mid Y) = 0.876$.

Answer

(a) $\theta \mid Y \sim \text{Beta}(3,9)$: mean 0.25, SD 0.120.
(b) Equal-tailed $(0.060, 0.518)$; HPD $(0.041, 0.484)$ — shorter and shifted toward 0 because of the right skew.
(c) $1 - \frac{990}{7980} = \mathbf{0.876}$.

Takeaway & common mistakes

In (c), do *not* plug in $\hat\theta = 0.25$: $1 - 0.75^{10} = 0.944$ overstates the probability, because it ignores uncertainty about θ. Averaging over the posterior (the PPD) gives the honest 0.876 — low values of θ make "all 10 die" quite likely.

Exercise 1.21 — Machine part failures (AND/OR systems)

Exercise 21 (exact text from the book)

X_1 and X_2 are binary indicators of failure for two parts of a machine. Independent tests have shown that $X_1 \sim \text{Bernoulli}(1/2)$ and $X_2 \sim \text{Bernoulli}(1/3)$. Y_1 and Y_2 are binary indicators of two system failures. We know that $Y_1 = 1$ if both $X_1 = 1$ and $X_2 = 1$ and $Y_1 = 0$ otherwise, and $Y_2 = 0$ if both $X_1 = 0$ and $X_2 = 0$ and $Y_2 = 1$ otherwise. Compute the following probabilities:

(a) The probability that $X_1 = 1$ and $X_2 = 1$ given $Y_1 = 1$.

(b) The probability that $X_1 = 1$ and $X_2 = 1$ given $Y_2 = 1$.

(c) The probability that $X_1 = 1$ given $Y_1 = 1$.

(d) The probability that $X_1 = 1$ given $Y_2 = 1$.

In plain words

Y_1 = "both parts failed" (AND). Y_2 = "at least one part failed" (OR). Work out what each system failure tells us about the parts.

Key idea

$Y_1 = X_1 X_2$ and $Y_2 = \max(X_1, X_2)$. Since Y is a deterministic function of (X_1, X_2): $\text{Prob}(A \mid Y = 1) = \text{Prob}(A \cap \{Y = 1\})/\text{Prob}(Y = 1)$.

Solution

Joint probabilities (independence):

x_1	x_2	prob	Y_1	Y_2
0	0	$\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$	0	0
1	0	$\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$	0	1
0	1	$\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$	0	1
1	1	$\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$	1	1

$\text{Prob}(Y_1 = 1) = \frac{1}{6}$ and $\text{Prob}(Y_2 = 1) = 1 - \frac{1}{3} = \frac{2}{3}$.

(a)

$Y_1 = 1$ happens *only* when both fail: $\text{Prob}(X_1 = X_2 = 1 \mid Y_1 = 1) = \frac{1/6}{1/6} = \mathbf{1}$.

(b)

$$\text{Prob}(X_1 = X_2 = 1 \mid Y_2 = 1) = \frac{1/6}{2/3} = \mathbf{\frac{1}{4}}.$$

(c)

$Y_1 = 1$ implies $X_1 = 1$: probability $\mathbf{1}$.

(d)

$$\text{Prob}(X_1 = 1 \mid Y_2 = 1) = \frac{\text{Prob}(X_1 = 1)}{\text{Prob}(Y_2 = 1)} = \frac{1/2}{2/3} = \mathbf{\frac{3}{4}}$$ (because $X_1 = 1$ implies $Y_2 = 1$).

Answer

(a) 1 (b) 1/4 (c) 1 (d) 3/4

Takeaway & common mistakes

An AND-failure is maximally informative (it pins down both parts). An OR-failure is only partially informative: it raises $\text{Prob}(X_1 = 1)$ from $\frac{1}{2}$ to $\frac{3}{4}$. Also note that, given $Y_2 = 1$, X_1 and X_2 become *dependent* ("explaining away"): if we learn $X_2 = 1$, X_1 drops back to $\frac{1}{2}$.

Exercise 1.22 — Clutch free throws in the NBA

Exercise 22 (exact text from the book)

The table below has the overall free throw proportion and results of free throws taken in pressure situations, defined as "clutch" (https://stats.nba.com/), for ten National Basketball Association players (those who received the most votes for the Most Valuable Player Award) for the 2016–2017 season. Since the overall proportion is computed using a large sample size, assume it is fixed, and analyze the clutch data for each player separately using Bayesian methods. Assume a uniform prior throughout this problem.

Player	Overall proportion	Clutch makes	Clutch attempts
Russell Westbrook	0.845	64	75
James Harden	0.847	72	95
Kawhi Leonard	0.880	55	63
LeBron James	0.674	27	39
Isaiah Thomas	0.909	75	83
Stephen Curry	0.898	24	26
Giannis Antetokounmpo	0.770	28	41
John Wall	0.801	66	82
Anthony Davis	0.802	40	54
Kevin Durant	0.875	13	16

(a) Describe your model for studying the clutch success probability including the likelihood and prior.

(b) Plot the posteriors of the clutch success probabilities.

(c) Summarize the posteriors in a table.

(d) Do you find evidence that any of the players have a different clutch percentage than overall percentage?

(e) Are the results sensitive to your prior? That is, do small changes in the prior lead to substantial changes in the posterior?

In plain words

For each player, estimate the probability of making a free throw under pressure and compare it to their (known) overall free-throw percentage.

Key idea

Beta–binomial model for each player separately; the key quantity for (d) is $\text{Prob}(\theta_i < q_i \mid Y_i)$ where q_i is the overall proportion. Values near 0 or 1 indicate evidence of a difference; values near 0.5 indicate none.

Solution

(a) Model.

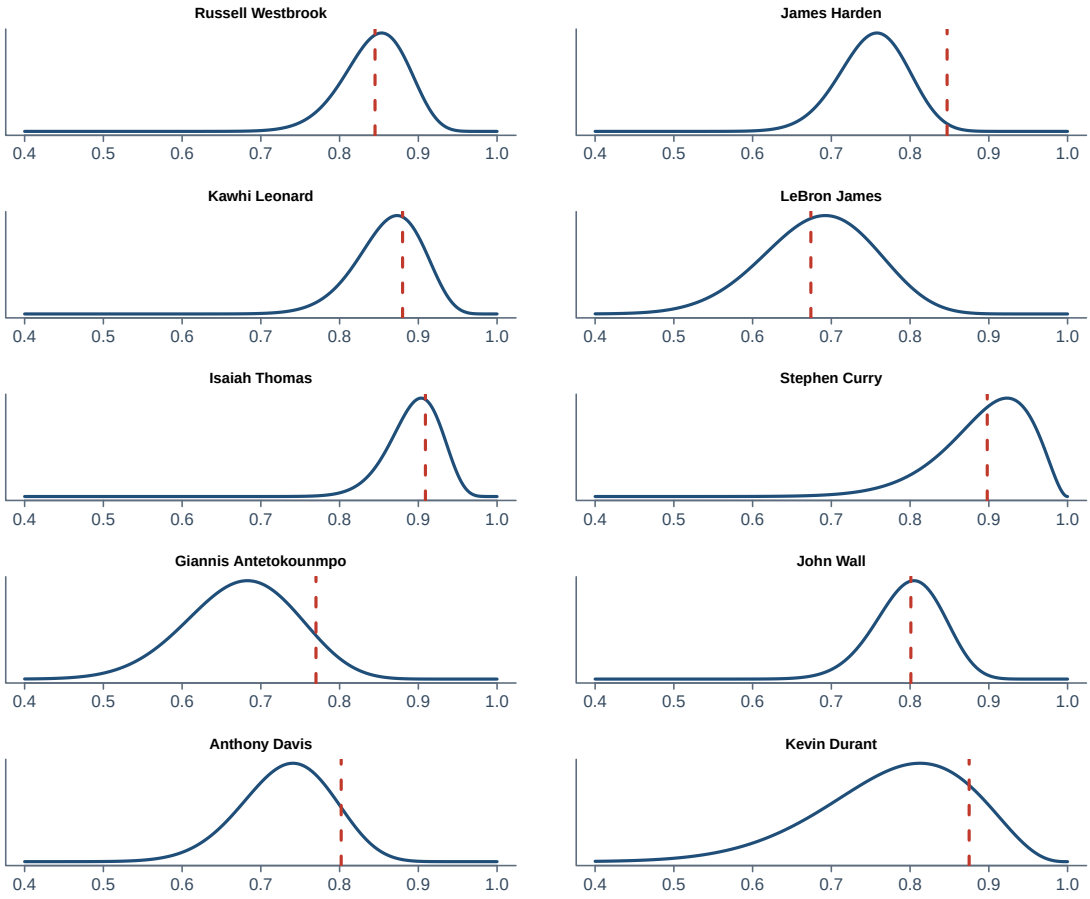
For player i with n_i clutch attempts and Y_i makes:

$$Y_i \mid \theta_i \sim \text{Binomial}(n_i, \theta_i), \qquad \theta_i \sim \text{Beta}(1,1) \quad \Longrightarrow \quad \theta_i \mid Y_i \sim \text{Beta}(Y_i + 1,\, n_i - Y_i + 1).$$

This assumes clutch shots are independent with a constant success probability for each player.

(b) Posteriors

(dashed red line: overall proportion q_i):



› (c) Summary table.

Player	Overall q_i	Posterior mean	95% interval	$\text{Prob}(\theta_i < q_i \mid Y_i)$
Russell Westbrook	0.845	0.844	(0.756, 0.916)	0.479
James Harden	0.847	0.753	(0.663, 0.833)	**0.991**
Kawhi Leonard	0.880	0.862	(0.768, 0.934)	0.640
LeBron James	0.674	0.683	(0.535, 0.814)	0.435
Isaiah Thomas	0.909	0.894	(0.821, 0.950)	0.645
Stephen Curry	0.898	0.893	(0.757, 0.976)	0.471
Giannis Antetokounmpo	0.770	0.674	(0.529, 0.804)	**0.917**
John Wall	0.801	0.798	(0.706, 0.876)	0.509
Anthony Davis	0.802	0.732	(0.610, 0.839)	0.887
Kevin Durant	0.875	0.778	(0.566, 0.932)	0.846

› (d) Evidence of a clutch effect?

- **James Harden**: $\text{Prob}(\theta < q) = 0.991$ and the 95% interval $(0.663, 0.833)$ excludes 0.847 — strong evidence he shoots *worse* in the clutch.
- **Giannis Antetokounmpo** (0.917) and **Anthony Davis** (0.887): moderate evidence of worse clutch shooting, but q is inside their 95% intervals.
- Everybody else: probabilities near 0.5 and intervals comfortably containing q — no evidence of a difference. No player shows evidence of being *better* in the clutch.

› (e) Prior sensitivity.

Repeat with Jeffreys $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ and $\text{Beta}(2, 2)$ priors:

Player	Uniform	Jeffreys	Beta(2,2)
Westbrook	0.479	0.435	0.565
Harden	0.991	0.989	0.994
Leonard	0.640	0.586	0.735
James	0.435	0.411	0.483
Thomas	0.645	0.588	0.745
Curry	0.471	0.366	0.657
Antetokounmpo	0.917	0.904	0.937
Wall	0.509	0.476	0.574
Davis	0.887	0.869	0.915
Durant	0.846	0.788	0.921

The *conclusions* do not change (Harden strong, Giannis/Davis moderate, others none). The numbers move most for players with few attempts (Curry $n = 26$: 0.37–0.66; Durant $n = 16$: 0.79–0.92), as expected — the less data, the more the prior matters.

Answer

(a) $Y_i|\theta_i \sim \text{Bin}(n_i, \theta_i)$, $\theta_i \sim \text{Beta}(1,1) \Rightarrow \text{Beta}(Y_i + 1, n_i - Y_i + 1)$. (b),(c) see figure and table. (d) Strong evidence only for Harden (worse in the clutch); moderate for Antetokounmpo and Davis. (e) Conclusions are robust to small prior changes; players with few attempts (Curry, Durant) are the most sensitive.

Takeaway & common mistakes

Looking at ten players and picking out the most extreme one is a *multiple comparisons* situation — one of ten looking unusual is not shocking. A hierarchical model that pools information across players (Chapter 7) would shrink the extreme estimates.

Exercise 1.23 — Who wrote the disputed Federalist Papers?

Exercise 23 (exact text from the book)

In the early twentieth century, it was generally agreed that Hamilton and Madison (ignore Jay for now) wrote 51 and 14 Federalist Papers, respectively. There was dispute over how to attribute 12 other papers between these two authors. In the 51 papers attributed to Hamilton, the word "upon" was used 3.24 times per 1,000 words, compared to 0.23 times per 1,000 words in the 14 papers attributed to Madison (for historical perspective on this problem, see [110]).

(a) If the word "upon" is used three times in a disputed text of length 1,000 words and we assume the prior probability 0.5, what is the posterior probability the paper was written by Hamilton?

(b) Give one assumption you are making in (a) that is likely unreasonable. Justify your answer.

(c) In (a), if we changed the number of instances of "upon" to one, do you expect the posterior probability to increase, decrease or stay the same? Why?

(d) In (a), if we changed the text length to 10,000 words and number of instances of "upon" to 30, do you expect the posterior probability to increase, decrease or stay the same? Why?

(e) Let Y be the number of observed number of instances of "upon" in 1,000 words. Compute the posterior probability the paper was written by Hamilton for each $Y \in \{0, 1, \ldots, 20\}$, plot these posterior probabilities versus Y and give a rule for the number of instances of "upon" needed before the paper should be attributed to Hamilton.

In plain words

Hamilton says "upon" a lot, Madison rarely. Use how often "upon" appears in a disputed paper to infer its author.

Key idea

Model word counts as Poisson: $Y \mid H \sim \text{Poisson}(3.24\,L)$ and $Y \mid M \sim \text{Poisson}(0.23\,L)$ for a text of L thousand words. Work with the **posterior odds** = prior odds × **likelihood ratio** (Bayes factor).

Solution

The likelihood ratio.

$$\frac{\text{Prob}(Y = y \mid H)}{\text{Prob}(Y = y \mid M)} = \frac{e^{-3.24L}(3.24L)^y/y!}{e^{-0.23L}(0.23L)^y/y!} = e^{-3.01L}\left(\frac{3.24}{0.23}\right)^y = e^{-3.01L}(14.087)^y.$$

(a) $y = 3$, $L = 1$, prior odds 1.

Likelihood ratio $= e^{-3.01} \times 14.087^3 = 0.04929 \times 2795.4 = 137.8$. So

$$\text{Prob}(H \mid Y = 3) = \frac{137.8}{1 + 137.8} = \mathbf{0.993}.$$

(b) An unreasonable assumption.

The Poisson model assumes "upon" occurrences are independent and occur at a constant rate throughout every paper. In reality word usage depends on topic and style; some papers will use "upon" in clusters, so counts are *over-dispersed* (variance > mean) — a negative binomial would be more realistic. (Also: treating 3.24 and 0.23 as known exactly ignores their estimation uncertainty, and a 0.5 prior ignores that Hamilton wrote more papers.)

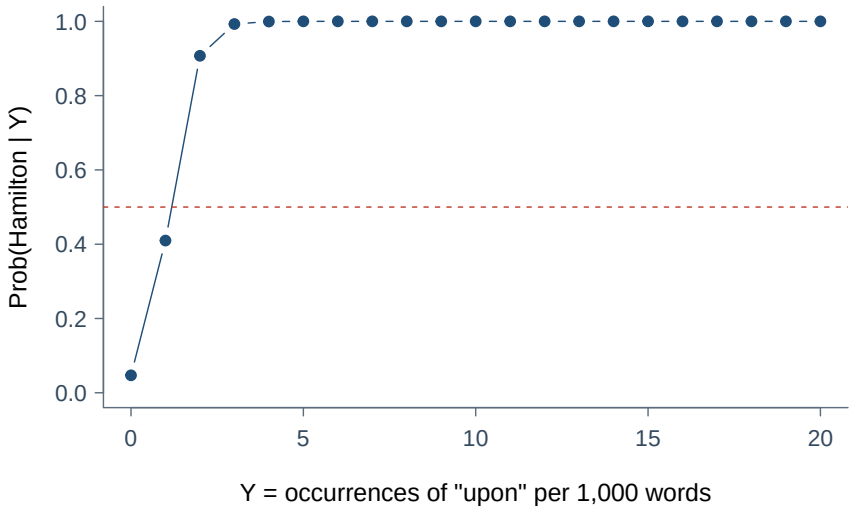
(c) $y = 1$: decrease.

One "upon" is more typical for Madison (expected 0.23) than for Hamilton (expected 3.24). LR $= 0.04929 \times 14.087 = 0.694 < 1$, so $\text{Prob}(H \mid Y = 1) = 0.410$ — a big *decrease*, now slightly favouring Madison.

(d) $L = 10$, $y = 30$: increase.

The *rate* is the same as in (a) (3 per 1,000), but we have ten times more text, so ten times more evidence: $\log \text{LR} = -30.1 + 30 \log 14.087 = 49.3$. The posterior rises to $1 - 4 \times 10^{-22} \approx 1$. More data with the same rate $\Rightarrow$ more certainty.

(e) The full curve.



Y	0	1	2	3	4	≥ 5
$\text{Prob}(H \mid Y)$	0.047	0.410	0.907	0.993	0.9995	> 0.9999

The posterior exceeds 0.5 when LR > 1, i.e. $y > 3.01/\log(14.087) = 1.14$.

Answer

(a) 0.993. (b) e.g. independent, constant-rate Poisson usage of "upon" (likely over-dispersed in reality). (c) Decrease, to 0.410. (d) Increase, to essentially 1.
(e) **Rule: attribute the paper to Hamilton if "upon" appears 2 or more times per 1,000 words** ($\text{Prob}(H \mid Y = 2) = 0.907$); for 99% confidence require $Y \geq 3$.

Takeaway & common mistakes

This is a miniature of Mosteller & Wallace's famous 1964 Bayesian study, which (using many words and negative binomial models) attributed all 12 disputed papers to Madison. The evidence scales with the amount of text, not just the rate.

Exercise 1.24 — Does the tutoring program work? Frequentist vs. Bayesian testing

Exercise 24 (exact text from the book)

Say $n = 1{,}000$ high school students are randomly selected to enter a tutorial program. It is known that 70% of the population from which they are drawn graduate from high school. After the program, it is found that $Y = 725$ of the n students graduate high school. Defining the true graduation rate for students in the program as $\theta \in (0, 1)$, we wish to test the null and alternative hypotheses

$$\mathcal{H}_o : \theta \leq 0.7 \text{ versus } \mathcal{H}_a : \theta > 0.7.$$

(a) Conduct a classical/frequentist test, and explain the results as if you are presenting them to a non-statistician.

(b) List a few subjective decisions in the analysis in (a).

(c) Conduct a Bayesian test, and explain the results as if you are presenting them to a non-statistician.

(d) Define a p-value and posterior probability of the null hypothesis, and describe how they are different.

In plain words

72.5% of tutored students graduated vs. 70% normally. Is that convincing evidence the program raises graduation, using (a) a classical test and (c) a Bayesian test?

Key idea

p-value: probability of data at least this extreme *if $\mathcal{H}_o$ were true*. **Bayesian**: compute $\text{Prob}(\mathcal{H}_o \mid \text{data}) = \text{Prob}(\theta \leq 0.7 \mid Y)$ directly from the posterior.

Solution

(a) Frequentist one-sided test.

Under $\theta = 0.7$ (the boundary of $\mathcal{H}_o$), $\hat{\theta} = 0.725$ has SE $\sqrt{0.7 \cdot 0.3/1000} = 0.01449$:

$$z = \frac{0.725 - 0.7}{0.01449} = 1.725, \qquad p = \text{Prob}(Z \geq 1.725) = 0.042.$$

(The exact binomial test gives $\text{Prob}(Y \geq 725 \mid \theta = 0.7) = 0.045$.) Since $p < 0.05$, we reject $\mathcal{H}_o$ at the 5% level.

For a non-statistician: "If the program had no benefit, getting 725 or more graduates out of 1,000 would happen only about 4 times in 100 by luck. That is unusual enough that we conclude the program probably helps — though the evidence is only modest."

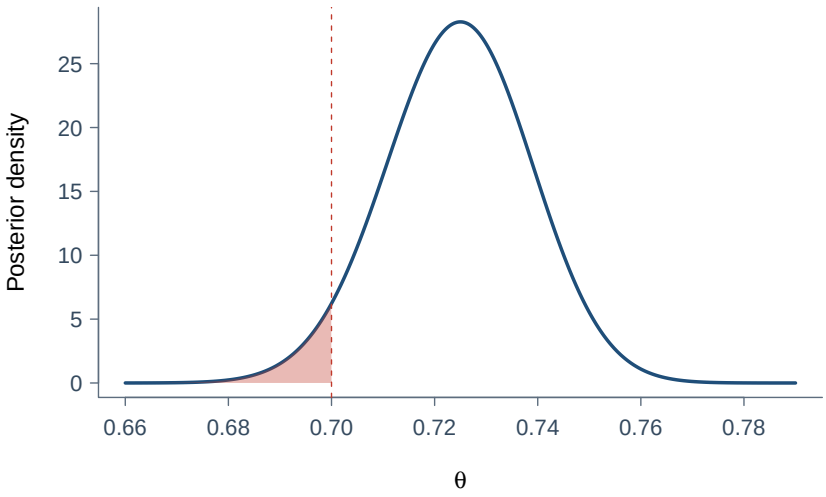
(b) Subjective decisions in (a).

- The significance level $\alpha = 0.05$ (why not 0.01 or 0.10?).
- One-sided vs. two-sided alternative.
- The test statistic and approximation (normal z vs. exact binomial vs. continuity correction) — these give $p = 0.042$ vs. 0.045.
- Treating the 70% population rate as known exactly, and assuming students are independent.

› (c) Bayesian test.

Uniform prior $\Rightarrow \theta \mid Y \sim \text{Beta}(726, 276)$, mean 0.7246, 95% interval $(0.697, 0.752)$.

$$\text{Prob}(\mathcal{H}_o \mid Y) = \text{Prob}(\theta \leq 0.7 \mid Y) = 0.043, \qquad \text{Prob}(\mathcal{H}_a \mid Y) = 0.957.$$



For a non-statistician: "Given these results, there is about a 96% chance that the program's graduation rate is above 70%. Our best estimate of the rate is 72.5%, and it is very likely between 69.7% and 75.2%."

Note: the uniform prior gives $\text{Prob}(\mathcal{H}_o) = 0.7$ *a priori*, so the Bayes factor for $\mathcal{H}_a$ is $\frac{0.957/0.043}{0.3/0.7} \approx 52$ — strong evidence.

› (d) Definitions and differences.

- **p-value** $= \text{Prob}(\text{data as or more extreme than observed} \mid \mathcal{H}_o \text{ true})$ — a statement about the *data*, assuming the hypothesis.
- **Posterior probability of** $\mathcal{H}_o = \text{Prob}(\mathcal{H}_o \text{ true} \mid \text{observed data})$ — a statement about the *hypothesis*, given the data.

They condition in opposite directions. The p-value is *not* the probability that $\mathcal{H}_o$ is true, though people often read it that way. Here they happen to agree numerically (0.042 vs. 0.043) — a known coincidence for one-sided tests with flat priors — but they can differ greatly for point nulls.

Answer

(a) $z = 1.73$, $p \approx 0.042$: reject $\mathcal{H}_o$ at 5%. (b) α, one-sided test, normal approximation, independence/known 70%.
(c) $\theta|Y \sim \text{Beta}(726, 276)$, $\text{Prob}(\mathcal{H}_o|Y) \approx 0.043$: about 96% probability the program helps. (d) See above: $\text{Prob}(\text{data} \mid \mathcal{H}_o)$ vs. $\text{Prob}(\mathcal{H}_o \mid \text{data})$.

Takeaway & common mistakes

"Statistically significant" $\neq$ "large effect": the estimated gain is 2.5 points, and the 95% interval reaches down to 69.7%. The Bayesian summary communicates both the direction and the size of the effect in plain language.

Exercise 1.25 — Moderna COVID-19 vaccine efficacy

Exercise 25 (exact text from the book)

The initial Moderna COVID-19 vaccine trial [9] produced the data in the table below

Placebo		Vaccine	
Infected	Participants	Infected	Participants
(table values and sub-questions missing from our copy of the book — see note)			

Note

In our copy of the book the exercise ends at the table header: the numbers and the sub-questions are missing. We use the published primary-analysis data of the trial (Baden et al., *New England Journal of Medicine* 384:403–416, 2021): **185 of 14,073** placebo and **11 of 14,134** vaccine participants had symptomatic COVID-19. The questions below are the natural ones for this chapter; if your copy differs, the same method applies — just change the four numbers in `ch01.R`.

In plain words

Compare the infection rates in the placebo and vaccine groups and estimate the vaccine efficacy $\text{VE} = 1 - \theta_V/\theta_P$ with a Bayesian analysis: (i) posteriors for the two infection rates, (ii) the posterior of VE with a 95% interval, (iii) the probability that VE exceeds 30% (the FDA's minimum) and 90%.

Key idea

Two independent beta–binomial models, then **Monte Carlo**: draw θ_P and θ_V from their posteriors and compute $1 - \theta_V/\theta_P$ for each draw. The draws of VE are a sample from its posterior — no algebra needed for the ratio.

Solution

(i) Infection rates.

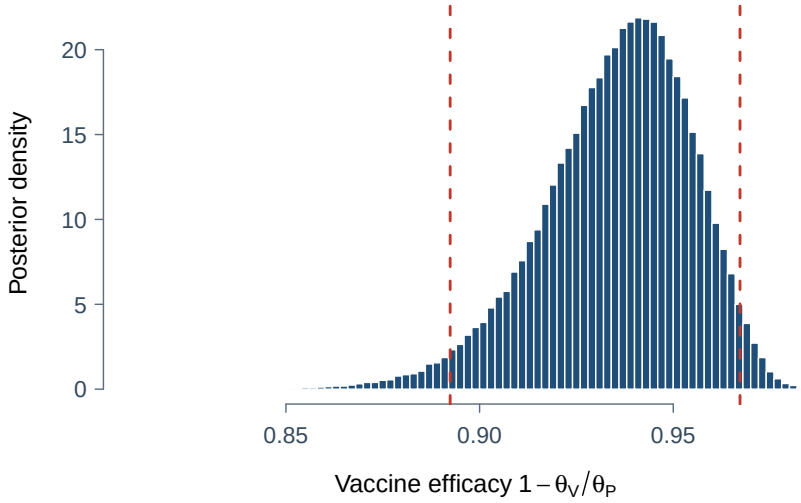
Uniform priors give

$$\theta_P \mid \text{data} \sim \text{Beta}(186,\ 13{,}889), \qquad \theta_V \mid \text{data} \sim \text{Beta}(12,\ 14{,}124),$$

with posterior means $\approx 1.32\%$ and 0.085%.

(ii) Efficacy by simulation.

```
thP ← rbeta(1e5, 1 + 185, 1 + 14073 - 185)
thV ← rbeta(1e5, 1 + 11,  1 + 14134 - 11)
VE  ← 1 - thV / thP
```



Posterior mean 0.935, median 0.938, 95% credible interval $(0.892,\ 0.967)$.

› (iii) Decision-relevant probabilities.

$\text{Prob}(\text{VE} > 0.3 \mid \text{data}) \approx 1$ (no draw out of 100,000 fell below), and $\text{Prob}(\text{VE} > 0.9 \mid \text{data}) \approx 0.95$.

Answer

VE posterior mean $\approx 93.5\%$, 95% interval $(89.2\%,\ 96.7\%)$; $\text{Prob}(\text{VE} > 30\%) \approx 1$, $\text{Prob}(\text{VE} > 90\%) \approx 0.95$. This matches the published estimate of 94.1% (95% CI 89.3–96.8%).

Takeaway & common mistakes

Monte Carlo makes inference on *any function* of parameters easy (ratios, differences, thresholds) — the main idea of Chapter 3. The tiny vaccine-arm count (11) dominates the uncertainty in VE.